Parity under standard-scaling neural-network Gaussian processes: superpolynomial and nearly exponential sample lower bounds at fixed depth

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Abstract

We study Bayesian learning of parity by fully connected networks with the standard neural-network Gaussian-process (NNGP) prior scaling: weight variances $\sigma_1^2/(d+1)$ in the first layer and σ_ℓ^2/N in later layers, at fixed depth, with width $N \to \infty$ first. The prior is conditioned on an output margin $|z(x)| > a$ at every Boolean input and then on the training signs; prediction counts as correct only when the correct label has posterior probability greater than $2/3$. For tanh networks of every fixed depth and every choice of layer gains, we prove that, for all large d, $n \le \exp(\frac{1}{20}\sqrt{d}\log d)$ uniform independent examples of full parity on d bits leave, with probability $1 - e^{-\Omega(d)}$ over the sample, at least a $1/16$ fraction of inputs without confident correct prediction. For the learner without the global margin, conditioned only on the training signs, a threshold of the same form $d^{\Theta(\sqrt{d})}$ (explicitly $d^{\sqrt{d}/10-1/2}/6000$ for $d \ge 25$) holds on every training set, for every bounded continuous nonconstant activation, every depth and every gain; the argument is kernel-agnostic and also gives the scale d^k for fixed k-sparse parity. On typical samples this learner fails far longer: for tanh at every fixed depth and gains, $n \le \exp(10^{-3} d \log\log d/\log d)$ examples leave, for all large d and with probability at least $3/4$, at least $3/8$ of the inputs without confident correct prediction. This is proved conditionally on the chaining estimate on which the collection's subcritical theorem also rests; the proof uses a random-restriction form of the Aspnes–Beigel–Furst–Rudich bound that depends only logarithmically on the amplitude of the posterior mean. An $\exp(\Omega(d))$ lower bound, the analogue of the subcritical theorem in this collection, remains open for both learners. For the sign-only learner the proof suggests that it would follow from two open log-free estimates at linear degree, for one-sided coercivity and for the sampled tail Gram matrix. For the margin learner we prove, again conditionally on that chaining estimate, that the step of the subcritical theorem which bounds the posterior's conditioning cost by a polynomial per sample is false at the standard scaling: for tanh networks, each fixed $\delta \in (0, 1/8)$ and $\exp(d^\epsilon) \le n \le e^{c_G(\delta) d}$, with probability at least $1 - 2\delta$ the cost per sample is superpolynomial in d. A sharper lower bound with the same status, of order a^2 over the prior tail variance per sample when $\log n \asymp d$, shows that no upper bound on that cost can match the first one, and that with our tail-Gram bound the head-bound argument of the subcritical paper, applied at the margin scale, cannot give an exponential bound whatever upper bound on the cost holds. Numerical spectra and small-dimensional posterior computations illustrate the results.

1 Introduction and results

The collection's subcritical theorem [1] proves an $\exp(\Omega(d))$ sample lower bound for full parity under a family of Gaussian network priors whose width scaling is indexed by an exponent $\gamma < 1$. In that family the hidden preactivations diverge with width, tanh acts as a sign function, and the limiting covariance is the iterated arcsine kernel. The usual NNGP scaling [7, 8, 9] is different: it keeps

the preactivations of order one, so the limit is the dual-activation kernel [6] of tanh itself. In the exponent notation of the collection's FAQ (Question 9) this scaling is the tuple $(0, 1, \dots, 1, 1)$, which is not on the one-parameter line $(\gamma - 1, \gamma, \dots, \gamma, \gamma + 1)$. The collection's NNGP note [2] proves a $\Theta(2^d)$ bound for a tanh NNGP whose depth grows linearly with d, and states that the fixed-depth problem is open.

We ask whether parity needs exponentially many samples for the fixed-depth standard-scaling learner. The answer we can prove is partial.

Results. All statements take width to infinity first, at fixed dimension, depth, gains, margin and training sample.

1. **Width limits** (Proposition 2.4, proved). For every bounded continuous nonconstant activation the network output converges to a nondegenerate Gaussian process with a dot-product kernel $K = A\,F((1 + x \cdot y)/(d + 1))$, where F has nonnegative Taylor coefficients that do not depend on d. Both learners' posterior label probabilities converge to those of the conditioned Gaussian process.

2. **Sign-only learner, every training set** (Theorem 3.2 and Corollary 3.4, proved). Without the global margin, for every such activation, every depth and every gain, every training set of size $n \le d^{\sqrt{d}/10 - 1/2}/6000$ (with $d \ge 25$; the explicit bound exceeds one only for $d \ge 385$) leaves, in the width limit, at least a quarter of the inputs of full parity without confidence above $2/3$. For k-sparse parity the threshold is $\binom{d}{\lfloor\sqrt{k}/5\rfloor}/6000$ (for $k \ge 25$), and for fixed k the failure fraction is at least $2^{-k} - O(n/\binom{d}{k})$. The same bounds hold for the margin learner when the readout variance is $\sigma_L^2 N^{-b}$ with $b < 1$ (Remark 3.5).

3. **Margin learner, typical samples** (Theorem 5.5, proved). For tanh at every fixed depth and every choice of gains and margin, $n \le \exp(\frac{1}{20}\sqrt{d}\log d)$ iid uniform examples of full parity leave, for $d \ge d_0$ (not explicit) and with probability at least $1 - e^{-d/10}$, Gibbs error at least $3/8$ and confidence failure at least $1/16$ in the width limit.

4. **Sign-only learner, typical samples** (Theorem 6.11). For tanh at every fixed depth and every choice of gains, $n \le \exp(10^{-3} d \log\log d / \log d)$ iid uniform examples of full parity leave, for $d \ge d_0$ and with probability at least $3/4$, at least $3/8$ of the inputs without confidence above $2/3$ in the width limit. This is proved conditionally on [1, Lemma 4.1], on which the subcritical theorem [1, Theorem 1.1] also rests, and unconditionally for $n \le n_0(d) = d^{\Theta(\sqrt{d})}$. The threshold d_0 is not explicit, and with our constants it is astronomically large (Remark 6.15). The same proof covers erf (Remark 6.17), and by Remark 3.5 the conclusion also holds for the margin learner with readout variance $\sigma_L^2 N^{-b}$, $b < 1$.

5. **The exponential question** (open). We do not know whether $\exp(\Omega(d))$ samples are needed. For the margin learner, Section 7 shows that one specific step of the subcritical proof fails. Its lower bounds on the conditioning cost (Propositions 7.2 and 7.3, Corollaries 7.4 and 7.5) are proved conditionally on [1, Lemma 4.1], like Theorem 6.11. That step bounds the conditioning cost per sample by a polynomial in d (at most $Cd^{1+\beta}$). Here, for each fixed $\delta \in (0, 1/8)$, with probability at least $1 - 2\delta$ and for $\exp(d^\epsilon) \le n \le e^{c_{\mathrm{G}}(\delta) d}$, where $c_{\mathrm{G}}(\delta) > 0$ is the constant of Lemma 7.1, that cost is superpolynomial in d (Proposition 7.2). A sharper bound (Proposition 7.3) puts the cost per sample above $c_\delta(a^2/\tau_r)\min\{1, A_\delta\sqrt{rH_r/d}\}$, which is of order a^2/τ_r at linear cutoffs; here τ_r is the prior variance above the adaptive cutoff degree r and $H_r = \log(e(d + 1)/r)$. So no upper bound matching the first one exists. With our

tail-Gram bound, the subcritical head estimate followed by the clipped-parity lemma at the margin scale gives nothing at linear cutoffs, whatever upper bound on the cost holds; we do not exclude the clipped-parity lemma at a larger amplitude scale (Section 7). For the sign-only learner, $\exp(\Omega(d))$ would follow from two open estimates, a log-free one-sided coercivity bound at linear degree and a log-free bound on the sampled tail Gram matrix (Remark 6.18; an extension suggested by the proof, not a result).

The sign-only result on every training set uses only a universal inequality between consecutive level masses of an isotropic kernel (Lemma 2.6). The margin result additionally needs a lower bound on every eigenvalue of the kernel. We derive it from an explicit Fourier-integral formula for the Hermite coefficients of tanh (Lemma 4.1). The typical-sample result needs no eigenvalue lower bound. It uses a two-step regularity $c_{m+2} \geq \eta c_m$ of the tanh dual coefficients, an intrinsic-dimension matrix Chernoff bound for the sampled tail Gram matrix, and a restriction lemma that bounds the agreement of a low-degree function with parity logarithmically, not linearly, in its amplitude (Section 6).

What this paper does not show. We prove no $\exp(\Omega(d))$ lower bound and no upper bound for either learner. We do not decide whether the exponents $\sqrt{d}\log d$ of the margin theorem and $d\log\log d/\log d$ of the typical-sample sign-only theorem are the truth. The margin theorem is proved for tanh, and for odd bounded continuous activations whose first-layer dual coefficients are all positive and decay at most like $e^{-o(\sqrt{m}\log m)}$, provided the linear coefficient $c_1(q_\ell)$ is positive at every later layer (Remark 5.6). It holds for $d \geq d_0$, where d_0 depends on the depth, gains and margin and is not explicit, because we prove no rate for the coefficient tail $T(s)$. It does not cover erf, whose coefficients decay geometrically; the typical-sample theorem does. Theorem 6.11 is proved for $n > n_0(d)$ conditionally on [1, Lemma 4.1], a bound for a squared Lipschitz process whose proof uses a homogeneous form of the intermediate estimate (2.20) in the proof of [10, Lemma 2.12], justified in [1, Remark 4.2]; we use that lemma without re-verifying it. The subcritical theorem [1, Theorem 1.1] and the lower bounds on the margin learner's conditioning cost in Section 7 (Propositions 7.2 and 7.3, Corollaries 7.4 and 7.5) rest on the same lemma. The threshold d_0 of Theorem 6.11 is not explicit and, with our constants, astronomically large (Remark 6.15), so the theorem is a purely asymptotic statement. The route to $\exp(\Omega(d))$ in Remark 6.18 assumes two estimates that are open, and the lower bound $\theta_r \geq c\, r^{-1/2}$ on the spectral ratio that enters the exact-norm part of Section 7 is only observed numerically, for the depth-2 tanh kernel (Section 8). As in [2, 1], the results concern the exact posterior; nothing is claimed about computing it.

Relation to kernel regression. The NNGP posterior mean given real-valued labels is kernel interpolation. For it, the span argument of [2, Proposition 4.1] and the known lower bounds for kernel methods on parities (references in [3]) concern squared error or correlation of the mean. Our learners condition on signs, or on signs and a global margin, and are judged by posterior confidence. That is a different criterion ([2, Section 5]; FAQ Question 7).

2 Model, width limits, and the cube spectrum

2.1 Network and learners

Let $\mathcal{X} = \{-1, 1\}^d$ with the uniform probability measure, $M = 2^d$ and $D = d + 1$. Fix a depth $L \geq 2$, an activation $\phi : \mathbb{R} \to \mathbb{R}$, gains $\sigma_1, \ldots, \sigma_L > 0$, and a margin $a > 0$. For width N put

$$h^0(x) = (1, x) \in \mathbb{R}^D, \qquad h^\ell(x) = \phi(W^\ell h^{\ell-1}(x)) \ (1 \leq \ell \leq L-1), \qquad z_N(x) = W^L h^{L-1}(x),$$

with independent centered Gaussian weights of variances

$$\operatorname{Var} W_{ij}^1 = \frac{\sigma_1^2}{D}, \qquad \operatorname{Var} W_{ij}^\ell = \frac{\sigma_\ell^2}{N} \ (2 \le \ell \le L-1), \qquad \operatorname{Var} W_i^L = \frac{\sigma_L^2}{N}. \tag{1}$$

All hidden layers have width N, and there are no biases besides the constant input coordinate. The *unit-gain* network has every $\sigma_\ell = 1$. The specified-gain family of [2, Section 7] has $\sigma_1 = 1$ and $\sigma_\ell^2 = 1/\mathbb{E}\tanh^2(G)$ for $\ell \ge 2$, with $G \sim N(0,1)$; at fixed depth it is another instance of (1).

Assumption 2.1. ϕ is bounded, continuous and not constant.

Let $f : \mathcal{X} \to \{-1, 1\}$ be a target and $S = (X_1, \dots, X_n)$ a sequence of training inputs. Define the events

$$B_S^{\mathrm{mar}} = \{z : \ |z(x)| > a \ \forall x \in \mathcal{X}, \ f(X_i)z(X_i) > 0 \ \forall i\},$$
$$B_S^{\mathrm{sgn}} = \{z : \ f(X_i)z(X_i) > 0 \ \forall i\}.$$

The *margin learner* is the prior conditioned on B_S^{mar}; this is the collection's convention (FAQ Question 1). The *sign-only learner* is the prior conditioned on B_S^{sgn}. For either learner put

$$q_{N,S}(x) = \mathbb{P}\{f(x)z_N(x) > 0 \mid z_N \in B_S\},$$
$$F_N(S) = \mathbb{P}_x\{q_{N,S}(x) \le 2/3\}, \qquad R_N(S) = \mathbb{E}_x[1 - q_{N,S}(x)].$$

F_N is the confidence-failure fraction and R_N the Gibbs error. For every posterior and every N,

$$R_N \le \frac{1 - F_N}{3} + F_N, \qquad \text{so} \qquad F_N \ge \frac{3R_N - 1}{2}, \tag{2}$$

since $1 - q \le 1$ on the failure set and $1 - q < 1/3$ off it.

2.2 Dual activations

For $q > 0$ write $v_\phi(q) = \mathbb{E}\phi(\sqrt{q}G)^2$ and, for $\rho \in [-1, 1]$,

$$\widehat{\phi}_q(\rho) = \frac{\mathbb{E}\phi(\sqrt{q}U)\phi(\sqrt{q}V)}{v_\phi(q)},$$

where (U, V) are standard Gaussians with correlation ρ. Under Assumption 2.1, $v_\phi(q) > 0$.

Lemma 2.2 (Dual activations)**.** *Expand $\phi(\sqrt{q}g) = \sum_{m \ge 0} \beta_m He_m(g)/\sqrt{m!}$ in $L^2(N(0,1))$, with probabilists' Hermite polynomials He_m. Then*

$$\widehat{\phi}_q(\rho) = \sum_{m \ge 0} c_m(q)\rho^m, \qquad c_m(q) = \frac{\beta_m^2}{\sum_k \beta_k^2} \ge 0, \qquad \sum_m c_m(q) = 1,$$

with absolute convergence on $[-1, 1]$. If ϕ is odd, $c_m(q) = 0$ for even m. Under Assumption 2.1, infinitely many $c_m(q)$ are positive.

Proof. For correlated standard Gaussians, $\mathbb{E}He_m(U)He_k(V) = \delta_{mk}\, m!\, \rho^m$; compare coefficients of $s^m t^k$ in $\mathbb{E}e^{sU - s^2/2}e^{tV - t^2/2} = e^{\rho st}$. The bilinear expansion of $\mathbb{E}\phi(\sqrt{q}U)\phi(\sqrt{q}V)$ then converges absolutely, because $|\rho| \le 1$ and $\sum \beta_m^2 = v_\phi(q) < \infty$. Odd functions have no even Hermite components. A bounded nonconstant function is not a polynomial, so infinitely many β_m are nonzero. □

Define $q_1 = \sigma_1^2$, $v_\ell = v_\phi(q_\ell)$, $q_{\ell+1} = \sigma_{\ell+1}^2 v_\ell$ for $1 \le \ell \le L-2$, and

$$A = \sigma_L^2 v_{L-1}, \qquad F = \widehat{\phi}_{q_{L-1}} \circ \cdots \circ \widehat{\phi}_{q_1}, \qquad K(x,y) = A\, F\!\left(\frac{1 + x \cdot y}{D}\right). \tag{3}$$

Lemma 2.3. *$F(t) = \sum_{m \ge 0} c_m t^m$ with $c_m \ge 0$ and $\sum_m c_m = 1$. The coefficients c_m and the scale A depend on ϕ, L and the gains, but not on d. If ϕ is odd, F is odd. Under Assumption 2.1, infinitely many c_m are positive.*

Proof. Composition of power series with nonnegative coefficients summing to one preserves both properties, and composition of odd series is odd. The variances q_ℓ do not involve d, because $\|h^0(x)\|^2 = D$ for every input. For the last claim, induct on the number of layers. Suppose H has infinitely many positive coefficients and $H(1) = 1$. Some $c_k(q)$ with $k \ge 1$ is positive; otherwise $\widehat{\phi}_q$ would be the constant $c_0(q) = 1$, forcing $\beta_m = 0$ for $m \ge 1$ and ϕ constant on the support of a Gaussian. The series H^{k-1} has some positive coefficient, at degree e say, so $H^k = H \cdot H^{k-1}$ has a positive coefficient at degree $e + m$ for each of the infinitely many m with $[t^m]H > 0$. □

2.3 Width limits

Proposition 2.4 (Width limits)**.** *Under Assumption 2.1, fix d, L and the gains.*

(i) As $N \to \infty$, z_N converges in law on $\mathbb{R}^{\mathcal{X}}$ to $G \sim N(0, K)$, with K as in (3).

(ii) K is positive definite (Lemma 2.5).

(iii) For every target f, training sequence S and margin $a > 0$, and for either learner, $\mathbb{P}\{z_N \in B_S\} \to \mathbb{P}\{G \in B_S\} > 0$, and $q_{N,S}(x) \to q_S(x) := \mathbb{P}\{f(x)G(x) > 0 \mid G \in B_S\}$ for every x. Consequently $R_N(S) \to R_S := \mathbb{E}_x[1 - q_S(x)]$, $\liminf_N F_N(S) \ge (3R_S - 1)/2$, and $\liminf_N F_N(S) \ge \mathbb{P}_x\{q_S(x) < 2/3\}$.

Proof. (i) For a positive semidefinite matrix Σ on $\mathcal{X}$ let $\Gamma(\Sigma) = \mathbb{E}[\phi(u)\phi(u)^\mathsf{T}]$ with $u \sim N(0, \Sigma)$. The map Γ is continuous on the positive semidefinite cone: if $\Sigma_k \to \Sigma$, then $\Sigma_k^{1/2}\xi \to \Sigma^{1/2}\xi$ for every ξ, and dominated convergence applies because ϕ is bounded and continuous. Write $Q_N^\ell(x,y) = N^{-1}\sum_i h_i^\ell(x) h_i^\ell(y)$ and let $\mathcal{F}_\ell$ be generated by the first ℓ layers of weights. The first-layer rows are independent, and each row's preactivation vector is $N(0, \Sigma_1)$ with $\Sigma_1(x,y) = \sigma_1^2(1 + x \cdot y)/D$. Conditionally on $\mathcal{F}_{\ell-1}$ with $\ell \ge 2$, the rows of the ℓth layer are independent with preactivation law $N(0, \sigma_\ell^2 Q_N^{\ell-1})$. Hence $\mathbb{E}[Q_N^\ell \mid \mathcal{F}_{\ell-1}] = \Gamma(\sigma_\ell^2 Q_N^{\ell-1})$ (with $\Gamma(\Sigma_1)$ at $\ell = 1$), and every entry has conditional variance at most $\|\phi\|_\infty^4/N$. By Chebyshev's inequality and the continuous mapping theorem, induction on ℓ gives $Q_N^\ell \to K^\ell$ in probability, where $K^1 = \Gamma(\Sigma_1)$ and $K^\ell = \Gamma(\sigma_\ell^2 K^{\ell-1})$. Every diagonal entry of $\sigma_\ell^2 K^{\ell-1}$ equals q_ℓ, and the correlations follow the recursion $\rho_1 = (1 + x \cdot y)/D$, $\rho_{\ell+1} = \widehat{\phi}_{q_\ell}(\rho_\ell)$. Therefore $\sigma_L^2 K^{L-1} = K$. Finally $z_N \mid \mathcal{F}_{L-1} \sim N(0, \sigma_L^2 Q_N^{L-1})$, so for each $t \in \mathbb{R}^{\mathcal{X}}$ the characteristic function is $\mathbb{E}\exp(-\frac{1}{2}\sigma_L^2 t^\mathsf{T} Q_N^{L-1} t)$, which converges to $\exp(-\frac{1}{2} t^\mathsf{T} K t)$ by bounded convergence.

(iii) Every coordinate $G(x)$ has variance $A > 0$. The boundary of B_S, and of $B_S \cap \{f(x)z(x) > 0\}$, lies in the finite union of hyperplanes $\{z(y) = \pm a\}$ and $\{z(y) = 0\}$, which is G-null. By the portmanteau theorem these probabilities converge. The limit of the denominator is positive because K is nonsingular and B_S is a nonempty open set. The last assertions follow from (2), from the finiteness of $\mathcal{X}$, and from the fact that $q_S(x) < 2/3$ forces $q_{N,S}(x) < 2/3$ for all large N. □

The quantifier order is the one in the main results: $N \to \infty$ at fixed d, L, gains, a and S. No quantitative width is claimed.

2.4 The kernel on the cube

For $V \subseteq \{0, 1, \dots, d\}$ write $\chi_V(u) = \prod_{i \in V} u_i$ on $\{-1, 1\}^D$, and χ_U for $U \subseteq [d]$ on $\mathcal{X}$. Let $p_m(j)$ be the probability that the walk on $\{-1, 1\}^D$ which flips one uniformly chosen coordinate per step is, after m steps, at a specified vertex at distance j from its start.

Lemma 2.5 (Cube spectrum; the kernel-free part of [1, Lemma 3.1])**.** *Let $F = \sum_m c_m t^m$ with $c_m \geq 0$ and $\sum c_m = 1$. Then $F(u \cdot u'/D) = \sum_V a_{|V|} \chi_V(u) \chi_V(u')$ on $\{-1, 1\}^D$, with $a_j = \sum_m c_m p_m(j)$, and:*

(a) $a_j \geq a_{j+2}$ for $0 \leq j \leq D - 2$;

(b) $a_j > 0$ if and only if $c_m > 0$ for some $m \geq j$ with $m \equiv j \pmod 2$;

(c) on $\mathcal{X}$, $K(x, y) = \sum_{U \subseteq [d]} \lambda_{|U|} \chi_U(x) \chi_U(y)$ with $\lambda_q = A(a_q + a_{q+1})$, and λ_q is nonincreasing in q;

(d) if infinitely many c_m are positive, every λ_q is positive and K is positive definite;

(e) if F is odd, $\lambda_{2k} = \lambda_{2k+1} = A a_{2k+1}$.

Proof. Expanding $(u \cdot u'/D)^m$ as an average over m independent uniform coordinates shows that its coefficient of $\chi_V(u)\chi_V(u')$ is $p_m(|V|)$. For (a), the one-step recursion $p_{m+1}(j) = \frac{j}{D} p_m(j-1) + \frac{D-j}{D} p_m(j+1)$ gives

$$D[p_{m+1}(j) - p_{m+1}(j+2)] = j[p_m(j-1) - p_m(j+1)] + (D-j-2)[p_m(j+1) - p_m(j+3)],$$

with $p_m(i) := 0$ for $i \notin [0, D]$, and induction from $m = 0$ gives $p_m(j) \geq p_m(j+2)$ (the case $j + 1 = D - 1$ of the hypothesis is $p_m(D-1) \geq p_m(D+1) = 0$, which is trivial); average against $c_m \geq 0$. For (b), $p_m(j) > 0$ exactly when $m \geq j$ and $m \equiv j$. For (c), restrict to $u = (1, x)$: $\chi_V(1, x) = \chi_{V \setminus \{0\}}(x)$, and each $U \subseteq [d]$ arises from $V = U$ and $V = U \cup \{0\}$. Monotonicity follows from (a): $\lambda_{q+1}/A = a_{q+1} + a_{q+2} \leq a_{q+1} + a_q$. For (d), given q there is $m \geq q + 1$ with $c_m > 0$, and by (b) one of a_q, a_{q+1} (the one whose index has the parity of m) is positive. The matrix K has eigenvalue $M\lambda_{|U|}$ on χ_U. Part (e) is (c) with $a_{\text{even}} = 0$. □

Throughout, λ_q denotes these *normalized* eigenvalues of K itself, so they include the factor A. For $g : \mathcal{X} \to \mathbb{R}$ write $\widehat{g}(U) = \mathbb{E}_x g(x)\chi_U(x)$, $\|g\|_2^2 = \mathbb{E}_x g(x)^2$, $g_{\leq r} = \sum_{|U| \leq r} \widehat{g}(U)\chi_U$ and $g_{>r} = g - g_{\leq r}$. The Gaussian energy is $\|g\|_K^2 = g^{\mathsf{T}} K^{-1} g = \sum_U \widehat{g}(U)^2/\lambda_{|U|}$, the convention of [1]. Let

$$\tau_r = \sum_{q > r} \binom{d}{q} \lambda_q = \operatorname{Var} G_{>r}(x), \qquad T(r) = \sum_{m > r} c_m,$$

and let $D_r = \sum_{q \leq r} \binom{d}{q}$ be the number of Walsh characters of degree at most r.

Lemma 2.6 (Level-mass inequalities)**.** *Let F be as in Lemma 2.5 and $0 \leq r < d$.*

(a) $\binom{d}{r+1} \lambda_{r+1} \leq \tau_r$.

(b) If F is odd and r is odd, $\binom{D}{r+2} \lambda_{r+1} \leq \tau_r$.

(c) $\tau_r \leq A\, T(r)$.

(d) $s_{2,r} := \sum_{q > r} \binom{d}{q} \lambda_q^2 \leq \lambda_{r+1} \tau_r$.

Proof. (a) τ_r contains the term $q = r+1$. (b) Here $\lambda_{r+1} = \lambda_{r+2}$ by Lemma 2.5(e), and $\binom{d}{r+1} + \binom{d}{r+2} = \binom{D}{r+2}$. (c) By the grouping in Lemma 2.5(c), $\tau_r/A = \sum_{V:|V\setminus\{0\}|>r} a_{|V|} \le \sum_{j>r}\binom{D}{j}a_j$, and $\binom{D}{j}a_j = \sum_m c_m \mathbb{P}\{\text{walk at distance } j \text{ at time } m\}$. Distance j at time m requires $m \ge j$, so $\sum_{j>r}\binom{D}{j}a_j \le \sum_{m>r} c_m$. (d) Bound one factor λ_q by λ_{r+1}, using monotonicity. □

Inequality (a) is what makes the sign-only argument kernel-agnostic. The subcritical paper uses the sharper arcsine ratio $\lambda_{r+1}/\tau_r \asymp 1/(r\binom{D}{r+2})$ [1, Lemma 3.2]; we use only (a) and (b), which hold for every such kernel, and in Section 6 a two-step regularity of the tanh level masses (Lemma 6.5).

3 The sign-only learner on every training set

In this section K is any positive-definite covariance on $\mathcal{X}$ that depends only on Hamming distance and whose degree eigenvalues λ_q are nonincreasing. By Lemma 2.5 this includes every kernel (3) under Assumption 2.1, for every depth and gain. Let $G \sim N(0, K)$ and

$$q_S(x) = \mathbb{P}\{f(x)G(x) > 0 \mid f(X_i)G(X_i) > 0 \ \forall i\}.$$

For $0 \le r < d$ let

$$\mathrm{err}_r(f) = \min_{\deg p \le r} \mathbb{P}_x\{f(x)p(x) \le 0\},$$

the least error of a degree-r polynomial threshold function with zeros counted as errors. A small constant shift of p removes its zeros without creating new errors, so this agrees with the definition used in [3].

Lemma 3.1 (Gaussian mean and confidence [1, Lemma 10.1])**.** *Let $\theta = 13/20$ and $c_\theta = \sqrt{2\pi}(e^{-1} - 7/20)$. Let $m = \mathbb{E}[G \mid f(X_i)G(X_i) > 0 \ \forall i]$, let $s \le n$ be the number of distinct training inputs, and let $\sigma_x^2 = \mathrm{Var}(G(x) \mid G_S)$. Then for every training sequence and every $0 \le v < d$:*

(i) $\|m\|_K^2 \le s$;

(ii) if $\sigma_x > 0$ *and* $q_S(x) > \theta$, *then* $f(x)m(x) > c_\theta \sigma_x$;

(iii) $\mathbb{P}_x\{\sigma_x^2 < \tau_v/2\} \le 2s\lambda_{v+1}/\tau_v \le 2n\lambda_{v+1}/\tau_v$;

(iv) $\|m - m_{\le v}\|_2^2 \le n\lambda_{v+1}$.

Proof. This is [1, Lemma 10.1]. Its proof uses Gaussianity, positive definiteness, Prékopa's theorem [14] and Fourier orthogonality. It uses the kernel only through two facts: the high-degree part $G_{>v}$ is independent of $G_{\le v}$ and has constant variance τ_v, and the largest normalized eigenvalue above degree v is λ_{v+1}. The second fact holds here by monotonicity. For (iii): conditioning also on $G_{\le v}$ can only lower the conditional variance, so $\sigma_x^2 \ge \tau_v - R(x,x)$, where $R = Q_{\mathcal{X}S}Q_{SS}^{\dagger}Q_{S\mathcal{X}}$ and Q is the covariance of $G_{>v}$. R is positive semidefinite of rank at most s, with $\mathrm{tr}\, R \le s\|Q\|_{\mathrm{op}} = sM\lambda_{v+1}$. Markov's inequality then gives (iii). Part (iv) follows from (i), since $\|m_{>v}\|_2^2 = \sum_{|U|>v}\widehat{m}(U)^2 \le \lambda_{v+1}\|m\|_K^2$. □

Theorem 3.2 (Sign-only learner)**.** *For every target f, every sequence S of n inputs (repeats and deliberately chosen inputs allowed), and every $0 \le r < d$,*

$$\mathbb{P}_x\{q_S(x) \le 13/20\} \ \ge\ \mathrm{err}_r(f) - C_0\frac{n\lambda_{r+1}}{\tau_r} \ \ge\ \mathrm{err}_r(f) - \frac{C_0\, n}{\binom{d}{r+1}}, \qquad C_0 = 2 + \frac{2}{c_\theta^2} < 1000. \quad (4)$$

If the eigenvalues are paired as in Lemma 2.5(e) and r is odd, $\binom{d}{r+1}$ may be replaced by $\binom{d+1}{r+2}$.

Proof. Let $p = m_{\le r}$. Let $E_1 = \{\sigma_x^2 < \tau_r/2\}$ and $E_2 = \{|m - p| \ge c_\theta\sqrt{\tau_r/2}\}$. By Lemma 3.1(iii), (iv) and Markov's inequality, $\mathbb{P}(E_1) + \mathbb{P}(E_2) \le (2 + 2/c_\theta^2)n\lambda_{r+1}/\tau_r$. Outside $E_1 \cup E_2$, the inequality $q_S(x) > 13/20$ implies $f(x)m(x) > c_\theta\sigma_x \ge c_\theta\sqrt{\tau_r/2} > |m(x) - p(x)|$, hence $f(x)p(x) > 0$. Therefore $\{fp \le 0\} \subseteq \{q_S \le 13/20\} \cup E_1 \cup E_2$. Since p has degree at most r, $\mathbb{P}_x\{fp \le 0\} \ge \mathrm{err}_r(f)$. The second inequality in (4) is Lemma 2.6(a), and the paired version is Lemma 2.6(b). Numerically $c_\theta = 0.04481\ldots$, so $C_0 = 997.7\ldots$. □

Lemma 3.3 (Parity is far from low-degree thresholds)**.** *Let $\chi_k(x) = x_1 \cdots x_k$ with $1 \le k \le d$, and let $0 \le s < k$. Then*

$$\mathrm{err}_s(\chi_k) \;\ge\; 2^{-k} \sum_{j \le (k-s-1)/2} \binom{k}{j} \;\ge\; \max\Big\{2^{-k},\; \frac{1}{2} - \frac{s+1}{\sqrt{2\pi k}}\Big\}.$$

Proof. On the k-cube the first inequality is the Aspnes–Beigel–Furst–Rudich bound [4], proved in [1, Lemma 6.2]. For $d > k$, fixing the coordinates outside $[k]$ turns a degree-s polynomial into a degree-s polynomial on the k-cube; average over the fixed coordinates. The term $j = 0$ gives 2^{-k}. Write $t = \lfloor (k - s - 1)/2 \rfloor$ and $B \sim \mathrm{Bin}(k, 1/2)$. By symmetry, $\mathbb{P}\{B \le t\} = \frac{1}{2}(1 - \mathbb{P}\{t < B < k - t\})$. The middle range contains $k - 2t - 1 \le s + 1$ integers, each of probability at most $\sqrt{2/(\pi k)}$. □

Corollary 3.4 (Networks without the global margin)**.** *Under Assumption 2.1, for every depth $L \ge 2$ and all gains, the sign-only network learner satisfies, for every target f, every training sequence S of length n and every $0 \le r < d$,*

$$\liminf_{N\to\infty} F_N(S) \;\ge\; \mathrm{err}_r(f) - \frac{C_0\, n}{\binom{d}{r+1}}.$$

In particular:

- *(a)* Full parity *($f = \chi_d$, $d \ge 25$): every training sequence with $n \le \binom{d}{\lfloor\sqrt{d}/5\rfloor}/(6C_0)$, and hence every one with $n \le d^{\sqrt{d}/10 - 1/2}/6000$, has $\liminf_N F_N(S) \ge 1/4$.*
- *(b)* Sparse parity *($f = \chi_k$, $25 \le k \le d$): every training sequence with $n \le \binom{d}{\lfloor\sqrt{k}/5\rfloor}/(6C_0)$ has $\liminf_N F_N(S) \ge 1/4$.*
- *(c)* Fixed k: $\liminf_N F_N(S) \ge 2^{-k} - C_0 n/\binom{d}{k}$ for every training sequence.*

These constants do not depend on L, the gains, or ϕ.

Proof. Combine Proposition 2.4(iii), the inclusion $\{q_S \le 13/20\} \subseteq \{q_S < 2/3\}$, and Theorem 3.2. For (a) and (b) take $r + 1 = \lfloor\sqrt{k}/5\rfloor$ (with $k = d$ in (a)). Lemma 3.3 gives $\mathrm{err}_r \ge \frac{1}{2} - \frac{1}{5\sqrt{2\pi}} > 0.42$, and the sample bound makes the subtracted term at most $1/6$. For the explicit form in (a), $\binom{d}{t} \ge (d/t)^t$ with $t = \lfloor\sqrt{d}/5\rfloor \ge \sqrt{d}/5 - 1$, so $\log\binom{d}{t} \ge (\sqrt{d}/5 - 1)\log(5\sqrt{d}) \ge \frac{\sqrt{d}}{10}\log d - \frac{1}{2}\log d$. For (c) take $r = k - 1$ and use $\mathrm{err}_{k-1}(\chi_k) \ge 2^{-k}$. □

Remark 3.5 (Larger readout variance). Corollary 3.4 also covers the *margin* learner when the readout variance is $\sigma_L^2 N^{-b}$ with $b < 1$, all other variances being as in (1). This is the tuple $(0, 1, \ldots, 1, b)$ in the notation of FAQ Question 9. Indeed $z_N = N^{(1-b)/2}\tilde{z}_N$, where $\tilde{z}_N$ is the output at $b = 1$. So the margin event is $\{|\tilde{z}_N(x)| > \varepsilon_N\ \forall x\}$ with $\varepsilon_N = aN^{-(1-b)/2} \to 0$. For fixed $\varepsilon > 0$ and large N, $B_\varepsilon \subseteq B_{\varepsilon_N} \subseteq B_S^{\mathrm{sgn}}$, where B_ε adds the condition $|z| > \varepsilon$ everywhere to the training signs. Proposition 2.4(i) and the portmanteau theorem bound the liminf and limsup of $\mathbb{P}\{\tilde{z}_N \in B_{\varepsilon_N}\}$ by $\mathbb{P}\{G \in B_\varepsilon\}$ and $\mathbb{P}\{G \in B_S^{\mathrm{sgn}}\}$. Letting $\varepsilon \downarrow 0$ shows that these agree in the limit,

since $\mathbb{P}\{G(x) = 0 \text{ for some } x\} = 0$. The same holds with the extra event $\{f(x)z(x) > 0\}$. So $q_{N,S}(x)$ converges to the sign-only $q_S(x)$, as in the $\gamma < 0$ case of [1, Proposition 2.2]. The same transfer applies to the typical-sample bound of Theorem 6.11.

Remark 3.6. (1) The bound holds on *every* training set, including deliberately chosen ones, and uniformly in depth, gain and activation. Depth may even grow with d. For comparison, [2, Theorem 7.1] proves a $\Theta(2^d)$ bound for a different learner, the margin learner with specified gains, at depth at least $C_a(d + \log(1/\epsilon_{\text{acc}}) + 1)$ for a fixed accuracy parameter ϵ_{acc}. (2) Theorem 3.2 is [3, Proposition 3] with the arcsine eigenvalue ratio replaced by Lemma 2.6(a). (3) The scale $\sqrt{d}\log d$ comes from requiring $\text{err}_r(f)$ to stay bounded below (the Aspnes–Beigel–Furst–Rudich lemma); with Lemma 2.6 alone, larger r yields at most a polynomially small failure fraction. It is not the limit for typical samples. For the arcsine kernel the subcritical paper improves $\sqrt{d}\log d$ to d [1, Theorem 10.2], using the sharp ratio $\lambda_{r+1}/\tau_r \asymp 1/(r\binom{D}{r+2})$ at r linear in d together with the clipped-parity lemma. A large improvement does not need the sharp ratio: for tanh, Theorem 6.11 (proved conditionally on [1, Lemma 4.1], and only for $d \geq d_0$ with d_0 astronomically large; Remarks 6.14 and 6.15) reaches $\log n$ of order $d \log\log d/\log d$ using only a two-step regularity of the dual coefficients, an intrinsic-dimension bound for the sampled tail Gram matrix and a restriction lemma that is logarithmic in the amplitude of the posterior mean (Section 6, Remark 6.16). For odd activations the paired form improves the binomial factor, for example to $\binom{d+1}{k+1}$ in Corollary 3.4(c) when k is even, without changing the exponent in (a) and (b).

4 Hermite coefficients of tanh and eigenvalue lower bounds

Lemma 4.1 (Hermite coefficients of tanh)**.** *Let $\sigma > 0$, $v = \mathbb{E}\tanh^2(\sigma G)$, and for odd m*

$$I_m(\sigma) = \int_0^\infty \frac{\omega^m e^{-\sigma^2\omega^2/2}}{\sinh(\pi\omega/2)}\, d\omega.$$

The dual coefficients of $\phi = \tanh$ at variance σ^2 are $c_m(\sigma^2) = \sigma^{2m} I_m(\sigma)^2/(m!\, v)$ for odd m and 0 for even m. In particular every odd coefficient is positive, and

$$c_m(\sigma^2) \;\geq\; \frac{2\sqrt{2\pi}}{\sigma^2 v\sqrt{m+1}}\, e^{-\pi\sqrt{m+1}/\sigma} \qquad (m \text{ odd}). \tag{5}$$

Proof. Gaussian integration by parts gives $\mathbb{E}[g(G)He_m(G)] = \mathbb{E}g^{(m)}(G)$ for smooth g with bounded derivatives. With $g(t) = \tanh(\sigma t)$ this gives $\beta_m = \sigma^m \mathbb{E}[\tanh^{(m)}(\sigma G)]/\sqrt{m!}$. The function $\text{sech}^2 = \tanh'$ has Fourier transform $\int_{\mathbb{R}} \text{sech}^2(y)e^{-i\omega y}dy = \pi\omega/\sinh(\pi\omega/2)$. This follows from the beta integral $\int_{\mathbb{R}} e^{-i\omega y}\,\text{sech}^2(y)dy = 2\Gamma(1+\frac{i\omega}{2})\Gamma(1-\frac{i\omega}{2})$ (substitute $u = (1+\tanh y)/2$, so $du = \frac{1}{2}\,\text{sech}^2\, y\, dy$ and $e^{2y} = u/(1-u)$) and $|\Gamma(1+ix)|^2 = \pi x/\sinh(\pi x)$; see [5, Sections 3.98 and 8.33]. The transform is positive, even and exponentially decaying. Fourier inversion and Fubini therefore give, for even $k = m-1$,

$$\mathbb{E}[\tanh^{(m)}(\sigma G)] = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{\pi\omega}{\sinh(\pi\omega/2)}(i\omega)^k e^{-\sigma^2\omega^2/2}d\omega = (-1)^{(m-1)/2} I_m(\sigma),$$

so $c_m = \beta_m^2/v = \sigma^{2m} I_m^2/(m!v)$. For the lower bound use $1/\sinh y \geq 2e^{-y}$. Let $J_m = \int_0^\infty \omega^m e^{-\sigma^2\omega^2/2}d\omega = \sigma^{-m-1}2^{(m-1)/2}\Gamma(\frac{m+1}{2})$, and let W have density proportional to $\omega^m e^{-\sigma^2\omega^2/2}$ on $(0,\infty)$. By Jensen's inequality, $I_m \geq 2J_m\mathbb{E}e^{-\pi W/2} \geq 2J_m e^{-\pi\mathbb{E}W/2}$. Here $\mathbb{E}W = J_{m+1}/J_m = \sqrt{2}\,\Gamma(\frac{m}{2}+1)/(\sigma\Gamma(\frac{m+1}{2})) \leq \sqrt{m+1}/\sigma$, by $\Gamma(x+\frac{1}{2}) \leq \sqrt{x}\,\Gamma(x)$. Finally the duplication formula gives $\sigma^{2m}J_m^2/m! = \sigma^{-2}\frac{\sqrt{\pi}}{2}\Gamma(\frac{m+1}{2})/\Gamma(\frac{m}{2}+1) \geq \sigma^{-2}\frac{\sqrt{\pi}}{2}\sqrt{2/(m+1)}$, again by $\Gamma(x+\frac{1}{2}) \leq \sqrt{x}\,\Gamma(x)$ with $x = \frac{m+1}{2}$. □

The formula is checked numerically in `numerics/spectrum.py`: the computed coefficients sum to one up to the truncation mass, and they reproduce a direct quadrature of the kernel (Section 8).

Corollary 4.2. *For $\phi = \tanh$ and every depth and gains, the composite coefficients of F satisfy, for every odd m,*

$$c_m \;\geq\; \kappa\, e^{-C_\phi \sqrt{m}}, \qquad \kappa = \frac{2\sqrt{2\pi}}{\sigma_1^2 v_1} \prod_{\ell=2}^{L-1} c_1(q_\ell) > 0, \qquad C_\phi = \sqrt{2}\Big(\frac{\pi}{\sigma_1} + 1\Big).$$

Proof. Since every $\widehat{\phi}_{q_\ell}$ has nonnegative coefficients, $\widehat{\phi}_{q_\ell}(u) \geq c_1(q_\ell) u$ coefficientwise, so $[t^m]F \geq \prod_{\ell \geq 2} c_1(q_\ell) \cdot c_m(q_1)$. Here $c_1(q) = q(\mathbb{E}\,\mathrm{sech}^2(\sqrt{q}G))^2 / v_\phi(q) > 0$. Apply (5) with $\sigma = \sigma_1$, and use $(m+1)^{-1/2} \geq e^{-\sqrt{m+1}}$ and $\sqrt{m+1} \leq \sqrt{2m}$. □

Lemma 4.3 (Eigenvalue lower bounds)**.** *Let F be odd with $c_m \geq \kappa e^{-\psi(m)}$ for every odd m, where $\kappa > 0$ and ψ is nondecreasing. For $d \geq 101$:*

(a) $\binom{d}{q}\lambda_q \;\geq\; \dfrac{A\kappa\, e^{-\psi(40D)}}{2D(D+1)}$ *for every $0 \leq q \leq 0.49d$;*

(b) the smallest eigenvalue of the matrix K satisfies $v_{\min}(K) \geq \frac{A\kappa}{8} e^{-\psi(\lceil 10D \log(2D) \rceil)}$.

Proof. Poissonize the walk. If the number of steps is $\Pi_t \sim \mathrm{Poisson}(t)$, written Π_t to avoid confusion with the width, the coordinates flip independently, each an odd number of times with probability $p(t) = (1 - e^{-2t/D})/2$. Hence $\sum_m \pi_t(m) p_m(j) = p(t)^j (1-p(t))^{D-j}$, where $\pi_t(m) = e^{-t} t^m / m!$. For odd j, only odd m contribute, and $p_m(j) \leq 1$ and $\pi_t(m) \leq 1$. Hence, for every integer m_1,

$$a_j \;\geq\; \Big(\min_{m \leq m_1,\ m \text{ odd}} c_m\Big)\Big[p(t)^j(1-p(t))^{D-j} - \mathbb{P}\{\Pi_t > m_1\}\Big], \tag{6}$$

and $\mathbb{P}\{\Pi_t > m_1\} \leq (et/m_1)^{m_1}$ whenever $et < m_1$, by the Chernoff bound $\mathbb{P}\{\Pi_t \geq m\} \leq e^{-t}(et/m)^m$ at $m = m_1 + 1$.

(a) Let $0 \leq q \leq 0.49d$ and let $j \in \{q, q+1\}$ be odd. Then $j \leq 0.49d + 1 \leq 0.495D$ because $D \geq 102$. Choose t with $p(t) = j/D$; then $t = \frac{D}{2} \log \frac{D}{D-2j} \leq \frac{D}{2} \log 100 < 2.31D$. Take $m_1 = 40D$. Then $\mathbb{P}\{\Pi_t > m_1\} \leq 0.158^{40D} < e^{-73D}$. The $\mathrm{Bin}(D, j/D)$ probability of its mode j is at least $1/(D+1)$, so (6) gives $\binom{D}{j} a_j \geq \kappa e^{-\psi(40D)}[\frac{1}{D+1} - 2^D e^{-73D}] \geq \kappa e^{-\psi(40D)}/(2(D+1))$. Since $\lambda_q = A a_j$ by Lemma 2.5(e), and $\binom{d}{q}/\binom{D}{j} \geq \min\{\frac{D-q}{D}, \frac{q+1}{D}\} \geq \frac{1}{D}$, part (a) follows.

(b) Let $j \leq D$ be odd, $t = \frac{D}{2}\log(2D)$ and $m_1 = \lceil 10D \log(2D) \rceil \geq 20t$. Then $p(t)^j(1-p(t))^{D-j} \geq 2^{-D}(1 - \frac{1}{2D})^D \geq 2^{-D-1}$, and $\mathbb{P}\{\Pi_t > m_1\} \leq (e/20)^{m_1} < 2^{-D}/4$. Thus $a_j \geq \kappa e^{-\psi(m_1)} 2^{-D}/4$. Every λ_q equals $A a_j$ for an odd $j \leq D$, and the matrix eigenvalues are $M\lambda_q = 2^{D-1}\lambda_q$. □

5 The margin learner

Let $G \sim N(0, K)$ with K as in (3), F odd, and

$$A_S = \{|G(x)| > a \ \forall x;\ y_i G(X_i) > 0 \ \forall i\}, \qquad \mathcal{D}(S, y) = -\log \mathbb{P}(A_S)$$

for labels $y \in \{-1, 1\}^n$. The posterior is the law of G given A_S; its relative entropy from $N(0, K)$ is exactly $\mathcal{D}(S, y)$. This is the $\gamma = 0$ structure of [1, Section 7], with the arcsine kernel replaced by K.

5.1 Conditioning cost

The bump functions below are those of [1, Lemma 7.2]; we repeat the parts of that proof that involve the kernel.

Lemma 5.1 (Globally margined interpolation)**.** *Suppose* $\varepsilon_d = \min_{q \le 0.49d} \binom{d}{q} \lambda_q > 0$, *and let* $v_{\min} = v_{\min}(K)$ *(the smallest eigenvalue of the matrix* K*). Put*

$$E_b = \frac{400}{(d+1)\varepsilon_d} + \frac{(d+1)e^{-0.0062d}}{v_{\min}}.$$

Let $X_1, \dots, X_n$ *be independent and uniform, with* $n(0.2)^{0.1d} \le 0.04$. *With probability at least* $1 - n^2 e^{-0.19d} - n\sqrt{A\,0.72^d / v_{\min}}$, *the following holds simultaneously for every labeling* y *and every* $A' > 0$. *There is* $h : \mathcal{X} \to \mathbb{R}$ *with* $|h(x)| > A'$ *for all* x, $y_i h(X_i) > A'$ *for all* i, *and*

$$\|h\|_K^2 \le 8A'^2\Big(\frac{1}{\lambda_0} + 8nE_b\Big).$$

Proof. For $z \in \mathcal{X}$ let $b_z(x) = 20\int_{2/5}^{9/20}(1-2p)^{\mathrm{dist}(x,z)}dp$. Then $b_z(z) = 1$, $0 \le b_z(x) \le (1/5)^{\mathrm{dist}(x,z)}$, and $\widehat{b_z}(U) = \chi_U(z)\widehat{b}_{|U|}$ with $\widehat{b}_j = 20\int_{2/5}^{9/20}(1-p)^{d-j}p^j dp$. Hence $\binom{d}{j}\widehat{b}_j \le 20/(d+1)$ and $\widehat{b}_j \le (3/5)^d$, and for $j > 0.49d$ the integrand increases in p, so $\widehat{b}_j \le 0.55^{d-j}0.45^j$ [1, proof of Lemma 7.2]. Therefore

$$\|b_z\|_K^2 = \sum_j \binom{d}{j}\frac{\widehat{b}_j^2}{\lambda_j} \le \sum_{j \le 0.49d} \frac{(20/(d+1))^2}{\varepsilon_d} + \frac{2^d}{v_{\min}} \sum_{j > 0.49d} \binom{d}{j} 0.55^{2(d-j)}0.45^{2j} \le E_b,$$

using $\lambda_j \ge v_{\min}/M$. For the last sum, the exponent $\log 2 + \mathsf{h}(t) + 2(1-t)\log 0.55 + 2t\log 0.45$, with h the binary entropy in nats, decreases for $t \ge 0.49$ and is below -0.0062 at $t = 0.49$, as computed in [1, proof of Lemma 7.2].

Let $J(z,w) = \langle b_z, b_w\rangle_K = \sum_U \nu_{|U|}\chi_U(z)\chi_U(w)$, with $\nu_j = \widehat{b}_j^2/\lambda_j \le 0.36^d M/v_{\min} = 0.72^d/v_{\min}$. For independent uniform X, Y, Parseval gives $\mathbb{E}J(X,Y)^2 = \sum_j \binom{d}{j}\nu_j^2 \le \nu_{\max}\|b_z\|_K^2$. Since $E_b \ge \|b_z\|_K^2 \ge b_z(z)^2/K(z,z) = 1/A$, Markov's inequality gives $\mathbb{P}\{\sum_{i\ne j}|J(X_i,X_j)| > nE_b\} \le n\sqrt{\nu_{\max}/E_b} \le n\sqrt{A\,0.72^d/v_{\min}}$. On the complement, every subset T of the indices satisfies $\|\sum_{i\in T} b_{X_i}\|_K^2 \le |T|E_b + \sum_{i\ne j}|J| \le 2nE_b$.

With probability at least $1 - n^2e^{-0.19d}$ all pairwise distances exceed $0.2d$, since $\log 2 - \mathsf{h}(0.2) > 0.19$. Then each x has at most one sample point within distance $0.1d$, so $\sum_{i\in T} b_{X_i}(x) \le 1 + n(0.2)^{0.1d}$, and the sum is at most $0.2 + 0.04$ unless $x = X_i$ with $i \in T$, where it is at least 1. Take $T = \{i : y_i = -1\}$ and $h = 2A'(1 - 2\sum_{i\in T} b_{X_i})$. Then $h \le -2A'$ on T and $h \ge 1.04A'$ elsewhere. Finally $\|h\|_K^2 \le 8A'^2(\|1\|_K^2 + 4\|\sum_T b\|_K^2)$, and $\|1\|_K^2 = 1/\lambda_0$. □

Lemma 5.2 (Conditioning cost)**.** *On the event of Lemma 5.1, for every labeling,*

$$\mathcal{D}(S,y) \le \log 2 + 4A'^2\Big(\frac{1}{\lambda_0} + 8nE_b\Big), \qquad A' = a + \sqrt{2A\log(4M)}.$$

Proof. Let $t_0 = \sqrt{2A\log(4M)}$ and $B = \{\max_x |G(x)| \le t_0\}$. A Gaussian tail and union bound give $\mathbb{P}(B) \ge 1/2$. With h from Lemma 5.1 at level $A' = a + t_0$, $h + B \subseteq A_S$. For a symmetric set B, the Cameron–Martin formula and $\cosh \ge 1$ give $\mathbb{P}(G \in h + B) \ge \mathbb{P}(B)e^{-\|h\|_K^2/2}$, as in [1, proof of Lemma 7.1]. □

5.2 Gibbs error

Lemma 5.3 (Gaussian quadratic entropy [1, Lemma 9.1])**.** *If Z is a linear image of a centered Gaussian vector, with covariance of trace t and largest eigenvalue ℓ, then $\mathbb{E}_Q\|Z\|^2 \le 2t + 4\ell D_0$ for every law Q of the underlying vector whose relative entropy from the Gaussian law is at most D_0.*

Proposition 5.4 (Gibbs error for every target)**.** *Let K be any positive-definite covariance on $\mathcal{X}$ that depends only on Hamming distance and has nonincreasing degree eigenvalues. For every target f, sample S and $0 \le s < d$, the posterior given A_S (labels $y_i = f(X_i)$) satisfies*

$$R_S := \mathbb{E}\big[\mathbb{P}_x\{f(x)G(x) < 0\} \mid A_S\big] \;\ge\; \mathrm{err}_s(f) - \frac{2\tau_s + 4\lambda_{s+1}\mathcal{D}(S,f)}{a^2}.$$

Proof. Apply Lemma 5.3 to $Z = M^{-1/2}G_{>s}$, whose covariance has trace τ_s and largest eigenvalue λ_{s+1}. This gives $\mathbb{E}[\|G_{>s}\|_2^2 \mid A_S] \le 2\tau_s + 4\lambda_{s+1}\mathcal{D}$. Fix a posterior draw and let $p = G_{\le s}$. Where $f(x)p(x) \le 0 < f(x)G(x)$ we have $|G - p| \ge fG = |G| > a$. Hence $\mathbb{P}_x\{fp \le 0\} \le \mathbb{P}_x\{fG < 0\} + \|G_{>s}\|_2^2/a^2$, and the left side is at least $\mathrm{err}_s(f)$. Take the posterior expectation. □

Theorem 5.5 (Margin learner, tanh)**.** *Let $\phi = \tanh$, and fix $L \ge 2$, gains $\sigma_1, \dots, \sigma_L > 0$ and a margin $a > 0$. There is d_0 such that for every $d \ge d_0$ and every integer $0 \le n \le \exp(\frac{1}{20}\sqrt{d}\log d)$ the following holds. With probability at least $1 - e^{-d/10}$ over independent uniform training inputs, the margin learner for full parity $f = \chi_d$ satisfies*

$$\lim_{N\to\infty} R_N(S) \ge \frac{3}{8}, \qquad \liminf_{N\to\infty} F_N(S) \ge \frac{1}{16}.$$

More generally, there is C, depending on L, the gains and a, such that on the same sample event, for every target f and every odd $s < d$,

$$\lim_{N\to\infty} R_N(S) \;\ge\; \mathrm{err}_s(f) - \frac{2A\,T(s)}{a^2}\Big(1 + \frac{2C(d^2 + n\,e^{C\sqrt{d}})}{\binom{d+1}{s+2}}\Big). \tag{7}$$

Proof. Spectral inputs. By Corollary 4.2, Lemma 4.3 applies with $\psi(m) = C_\phi\sqrt{m}$. Thus $\varepsilon_d \ge e^{-C\sqrt{d}}$ and $v_{\min} \ge e^{-C\sqrt{d\log d}}$ for large d, where here and below C denotes constants depending only on L, the gains and a. In particular $E_b \le e^{C\sqrt{d}}$. Also $\lambda_0 = Aa_1 \ge Ac_1/D$, since $p_1(1) = 1/D$.

Sample event. Let $n \le \exp(\frac{1}{20}\sqrt{d}\log d) = e^{o(d)}$. Then $n(0.2)^{0.1d} \le 0.04$, and the failure probability in Lemma 5.1 is at most $e^{-0.18d} + e^{-0.16d + C\sqrt{d\log d}} \le e^{-d/10}$ for large d. On this event Lemma 5.2 and $A'^2 \le Cd$ give, for every labeling, $\mathcal{D}(S,y) \le C(d^2 + ne^{C\sqrt{d}})$.

Every target. Combine Proposition 5.4 with Lemma 2.6(b),(c), which give $\lambda_{s+1} \le \tau_s/\binom{D}{s+2}$ and $\tau_s \le AT(s)$. Then Proposition 2.4(iii) gives (7).

Parity. Let s be the largest odd integer with $s+1 \le \sqrt{d}/5$. Lemma 3.3 gives $\mathrm{err}_s(\chi_d) > 0.42$. Since $s+2 \ge \sqrt{d}/5 - 1$, $\log\binom{D}{s+2} \ge (s+2)\log\frac{D}{s+2} \ge \frac{1}{10}\sqrt{d}\log d - \log d$. For $n \le \exp(\frac{1}{20}\sqrt{d}\log d)$ the fraction in (7) is therefore at most $\exp(-\frac{1}{20}\sqrt{d}\log d + C\sqrt{d} + C\log d) \to 0$. Also $T(s) \to 0$ as $d \to \infty$, because $s \to \infty$ and T does not depend on d (Lemma 2.3). Hence $\lim_N R_N(S) \ge 0.42 - o(1) \ge 3/8$ for large d, and (2) gives the bound on F_N. □

Remark 5.6 (Thresholds, constants, activations). (1) Taking $s+1 \le \delta\sqrt{d}$ in the proof gives $\lim_N R_N \ge \frac{1}{2} - \frac{\delta}{\sqrt{2\pi}} - o(1)$ for $n \le \exp(\frac{\delta}{4}\sqrt{d}\log d)$. For $q > 1/2$ take any $\delta < \sqrt{2\pi}(q - \frac{1}{2})$ and $c_q = \delta/4$; the inequality below holds at every width and passes to the limit since $R_N \to R_S$. By the second inequality in FAQ Question 5, $F_q \ge (R - (1-q))/q$, so for each fixed $q > 1/2$ the fraction of inputs

with correct-label probability at most q stays bounded below in a range $n \le \exp(c_q\sqrt{d}\log d)$. (2) The constant $\frac{1}{20}$ is not optimized. (3) The proof uses tanh only through oddness, Assumption 2.1, and the lower bound of Corollary 4.2. It works verbatim for any odd activation satisfying Assumption 2.1 whose composite coefficients satisfy $c_m \ge \kappa e^{-\psi(m)}$ for odd m (by the argument of Corollary 4.2 it suffices that the first-layer coefficients satisfy this and that $c_1(q_\ell) > 0$ for $\ell \ge 2$), with ψ nondecreasing and $\psi(m) = o(\sqrt{m}\log m)$. Then $E_b = e^{o(\sqrt{d}\log d)}$, $v_{\min} \ge e^{-o(d)}$, and the same threshold results. The error function is not covered: since $\mathbb{E}\,\mathrm{erf}(\sigma U)\,\mathrm{erf}(\sigma V) = \frac{2}{\pi}\arcsin(\frac{2\sigma^2\rho}{1+2\sigma^2})$, its dual coefficients decay geometrically in m. (4) For k-sparse parity the same proof needs $e^{C\sqrt{d}} \ll \binom{D}{\delta\sqrt{k}}$, roughly $k \ge Cd/\log^2 d$; we have not pursued smaller k.

6 The sign-only learner on typical samples: a nearly exponential bound

Theorem 3.2 holds on every training set, and its scale $d^{\Theta(\sqrt{d})}$ comes from requiring $\mathrm{err}_r(f)$ to stay bounded below. For independent uniform training inputs we go much further. In this section $f = \chi_{[d]}$ is full parity, $G \sim N(0, K)$, and, as in Section 3,

$$q_S(x) = \mathbb{P}\{f(x)G(x) > 0 \mid f(X_i)G(X_i) > 0\ \forall i\}, \qquad m = \mathbb{E}[G \mid f(X_i)G(X_i) > 0\ \forall i],$$

and $F_\theta(S) = \mathbb{P}_x\{q_S(x) \le \theta\}$, with $\theta = 13/20$ and c_θ as in Lemma 3.1. The main result of this section, Theorem 6.11, says that for tanh at every fixed depth and gains, $n \le \exp(10^{-3}d\log\log d/\log d)$ examples leave $F_\theta(S) \ge 3/8$ with probability at least $3/4$ over the sample. The proof follows the skeleton of [1, Theorem 10.2], with three changes. The clipped-parity lemma, which is linear in the amplitude of the posterior mean, is replaced by a restriction lemma that is logarithmic in it (Section 6.1). The sharp arcsine level estimates are replaced by a two-step regularity of the tanh dual coefficients (Section 6.2). The far-pair analysis of the sampled tail Gram matrix is replaced by an intrinsic-dimension matrix Chernoff bound (Section 6.3).

Status. For $n \le n_0(d)$ (defined in Section 6.5) the theorem follows from Theorem 3.2 and is unconditional. For larger n the proof uses the squared coercivity lemma [1, Lemma 4.4], whose proof rests on [1, Lemma 4.1]. That lemma is a bound for a squared Lipschitz process; its proof uses a homogeneous form of the intermediate estimate (2.20) in the proof of Lee's chaining theorem [10, Lemma 2.12], justified in [1, Remark 4.2]. We use [1, Lemma 4.1] as a black box. So Theorem 6.11 is proved conditionally on [1, Lemma 4.1], the lemma on which the subcritical theorem [1, Theorem 1.1] also rests (Remark 6.14).

6.1 Parity after a random restriction

For integers $j, k \ge 0$ let

$$\mathrm{agr}_j(k) = \max_{\deg P \le j} \mathbb{P}_{y\in\{-1,1\}^k}\{\chi_{[k]}(y)P(y) > 0\} = 1 - \mathrm{err}_j(\chi_{[k]}),$$

with the error computed on the k-cube. Thus $\mathrm{agr}_j(k) = 1$ if $j \ge k$, in particular for $k = 0$, and Lemma 3.3, applied on the k-cube to $\chi_{[k]}$, gives

$$\mathrm{agr}_j(k) \;\le\; 1 - 2^{-k} \sum_{i \le (k-j-1)/2} \binom{k}{i} \;\le\; \frac{1}{2} + \frac{j+1}{\sqrt{2\pi k}} \qquad (0 \le j < k). \tag{8}$$

Lemma 6.1 (Restriction lemma)**.** *For every $h : \mathcal{X} \to \mathbb{R}$, every $p \in (0,1]$, every integer $j \geq 0$ and every $b > 0$,*

$$\mathbb{P}_x\{f(x)h(x) \geq b\} \;\leq\; \mathbb{E}\,\mathrm{agr}_j(|\mathcal{I}|) + \frac{1}{b^2}\sum_{U\subseteq[d]} \widehat{h}(U)^2\,\mathbb{P}\{|U\cap\mathcal{I}| > j\}, \tag{9}$$

where the random set $\mathcal{I} \subseteq [d]$ contains each coordinate independently with probability p. No margin is required off the event $\{fh \geq b\}$, and the constant term $\widehat{h}(\emptyset)$ carries weight zero.

Proof. Fix a realization I of $\mathcal{I}$ and write $x = (y,z)$ with $y \in \{-1,1\}^I$ and $z \in \{-1,1\}^{[d]\setminus I}$; the uniform measure on $\mathcal{X}$ is the product of the uniform measures on y and z. For fixed z, the function $h_z(y) = h(y,z)$ has Walsh coefficients $\widehat{h_z}(T) = \sum_{U:U\cap I=T}\widehat{h}(U)\chi_{U\setminus I}(z)$, $T \subseteq I$. Split $h_z = h_z^{\mathrm{lo}} + h_z^{\mathrm{hi}}$, where h_z^{lo} keeps the coefficients with $|T| \leq j$. Parity restricts to $f(y,z) = \varepsilon_z\chi_I(y)$ with $\varepsilon_z = \chi_{[d]\setminus I}(z) \in \{\pm 1\}$. Where $fh \geq b$ and $|h_z^{\mathrm{hi}}| < b$, we have $fh_z^{\mathrm{lo}} = fh - fh_z^{\mathrm{hi}} > 0$. Hence

$$\mathbf{1}\{fh \geq b\} \leq \mathbf{1}\{\chi_I\,\varepsilon_z h_z^{\mathrm{lo}} > 0\} + \mathbf{1}\{|h_z^{\mathrm{hi}}| \geq b\}.$$

Since $\varepsilon_z h_z^{\mathrm{lo}}$ has degree at most j on the $|I|$-cube, the first indicator has mean at most $\mathrm{agr}_j(|I|)$ over y. By Markov's inequality the second has mean at most $b^{-2}\sum_{T\subseteq I,\,|T|>j}\widehat{h_z}(T)^2$ over y. Two distinct sets U, U' with $U\cap I = U'\cap I$ have $U\setminus I \neq U'\setminus I$, so the characters $\chi_{U\setminus I}$ in the sum defining $\widehat{h_z}(T)$ are orthonormal in z, and $\mathbb{E}_z\widehat{h_z}(T)^2 = \sum_{U\cap I=T}\widehat{h}(U)^2$. Therefore $\mathbb{P}_x\{fh \geq b\} \leq \mathrm{agr}_j(|I|) + b^{-2}\sum_U \widehat{h}(U)^2\,\mathbf{1}\{|U\cap I| > j\}$. Average over $\mathcal{I}$. □

Corollary 6.2 (Explicit form)**.** *Let $1 \leq u \leq d$, let h have degree at most u, let $b > 0$, and put $\rho_u = u/d$ and*

$$\mathcal{R} = \frac{1}{b^2}\sum_{U\neq\emptyset}\widehat{h}(U)^2.$$

For every integer J with $1 \leq J \leq e^2u$,

$$\mathbb{P}_x\{fh \geq b\} \;\leq\; \frac{1}{2} + e\sqrt{\frac{J\rho_u}{\pi}} + \frac{1}{2}\exp\Bigl(-\frac{J}{8e^2\rho_u}\Bigr) + \mathcal{R}\,e^{-J}. \tag{10}$$

In particular, if $\varepsilon \in (0,1]$ and $1 + \log_+(\mathcal{R}/\varepsilon) \leq e^2u$, then

$$\mathbb{P}_x\{fh \geq b\} \leq \frac{1}{2} + \varepsilon + e\sqrt{\frac{\rho_u\,(1+\log_+(\mathcal{R}/\varepsilon))}{\pi}} + \frac{1}{2}e^{-1/(8e^2\rho_u)}.$$

Proof. Take $j = J-1$ and $p = J/(e^2u) \in (0,1]$ in (9). Then $|\mathcal{I}| \sim \mathrm{Bin}(d,p)$ has mean $dp = J/(e^2\rho_u)$. By (8), $\mathrm{agr}_j(k) \leq \frac{1}{2} + J/\sqrt{2\pi k}$ for $k \geq 1$ (trivially if $j \geq k$), and $\mathrm{agr}_j \leq 1$ always. Split according to whether $|\mathcal{I}| \geq dp/2$ and use the Chernoff bound $\mathbb{P}\{|\mathcal{I}| < dp/2\} \leq e^{-dp/8}$:

$$\mathbb{E}\,\mathrm{agr}_j(|\mathcal{I}|) \leq \frac{1}{2} + \frac{J}{\sqrt{\pi dp}} + \frac{1}{2}e^{-dp/8} = \frac{1}{2} + e\sqrt{\frac{J\rho_u}{\pi}} + \frac{1}{2}e^{-J/(8e^2\rho_u)}.$$

In the second term of (9), $\widehat{h}(U) = 0$ for $|U| > u$, the weight vanishes for $U = \emptyset$, and for $1 \leq |U| \leq u$ a union bound over J-subsets of U gives $\mathbb{P}\{|U\cap\mathcal{I}| \geq J\} \leq \binom{u}{J}p^J \leq (eup/J)^J = e^{-J}$ (the probability is zero if $J > u$). For the last claim take $J = \max\{1, \lceil\log(\mathcal{R}/\varepsilon)\rceil\} \leq 1 + \log_+(\mathcal{R}/\varepsilon)$, so that $\mathcal{R}e^{-J} \leq \varepsilon$, and use $J \geq 1$ in the exponential. □

With $p = 1$ and $\deg h \leq j$, Lemma 6.1 reduces to (8) on the whole cube. Remark 6.16 compares Corollary 6.2 with the clipped-parity lemma.

6.2 Two-step regularity of the dual coefficients

Lemma 6.3 (First layer). *Let $\sigma > 0$, $\varpi = \pi/(2\sigma)$ and $\upsilon_m = \frac{1}{2}(\sqrt{\varpi^2 + 4m} - \varpi)$. The dual coefficients of* tanh *at variance σ^2 satisfy, for every odd $m \geq 1$,*

$$\eta(\sigma) := \frac{1}{6(1+\varpi)^4} \;\leq\; \frac{\upsilon_m^4}{(m+1)(m+2)} \;\leq\; \frac{c_{m+2}(\sigma^2)}{c_m(\sigma^2)} \;\leq\; \frac{m^2}{(m+1)(m+2)}.$$

Proof. By Lemma 4.1, $c_{m+2}/c_m = \sigma^4 (I_{m+2}/I_m)^2/((m+1)(m+2)) = (\sigma^2 \mathbb{E}_\mu \omega^2)^2/((m+1)(m+2))$, where μ is the probability measure on $(0,\infty)$ with density proportional to $g(\omega) = \omega^m e^{-\sigma^2\omega^2/2}/\sinh(\pi\omega/2)$. Since $\omega g(\omega) \sim \frac{2}{\pi}\omega^m \to 0$ at 0 and g has Gaussian decay, integration by parts gives $\int_0^\infty \omega g'(\omega)\, d\omega = -\int_0^\infty g(\omega)\, d\omega$. As $\omega g'/g = m - \sigma^2\omega^2 - \frac{\pi\omega}{2}\coth\frac{\pi\omega}{2}$, this reads

$$\sigma^2 \mathbb{E}_\mu \omega^2 = m + 1 - \mathbb{E}_\mu\Big[\frac{\pi\omega}{2}\coth\frac{\pi\omega}{2}\Big].$$

Since $y\coth y \geq 1$, $\sigma^2\mathbb{E}_\mu\omega^2 \leq m$, which gives the upper bound. Since $y\coth y = y + 2y/(e^{2y}-1) \leq 1 + y$, Jensen's inequality gives $\sigma^2\mathbb{E}_\mu\omega^2 \geq m - \frac{\pi}{2}(\mathbb{E}_\mu\omega^2)^{1/2}$. With $Y = \sigma(\mathbb{E}_\mu\omega^2)^{1/2}$ this is $Y^2 + \varpi Y - m \geq 0$, so $Y \geq \upsilon_m = 2m/(\varpi + \sqrt{\varpi^2+4m}) \geq m/(\varpi + \sqrt{m})$, and $c_{m+2}/c_m = Y^4/((m+1)(m+2)) \geq \upsilon_m^4/((m+1)(m+2))$. Finally, for $m \geq 1$, $\varpi + \sqrt{m} \leq (1+\varpi)\sqrt{m}$ and $(m+1)(m+2) \leq 6m^2$, so $\upsilon_m^4/((m+1)(m+2)) \geq m^4/((1+\varpi)^4 m^2 \cdot 6m^2) = \eta(\sigma)$. □

Lemma 6.4 (Composites). *Let $H(t) = \sum_{m\geq 1} h_m t^m$ have nonnegative coefficients with $h_{m+2} \geq \eta h_m$ for all $m \geq 1$, and let $g_0, g_1, \ldots \geq 0$. Then the coefficients of $\sum_k g_k H^k$ satisfy $[t^{m+2}] \geq \eta\,[t^m]$ for all $m \geq 1$. Consequently, for $\phi =$ tanh, every depth and all gains, with $\eta = \eta(\sigma_1)$ and $c_1 = \prod_{\ell=1}^{L-1} c_1(q_\ell) > 0$ the linear coefficient of F,*

$$c_{m+2} \geq \eta\, c_m \qquad \text{and} \qquad c_m \geq \eta^{(m-1)/2} c_1 \qquad (m \text{ odd}). \tag{11}$$

Proof. For $k \geq 1$, $[t^m]H^k = \sum h_{j_1}\cdots h_{j_k}$ over compositions $j_1 + \cdots + j_k = m$ with all $j_i \geq 1$. The map $(j_1, \ldots, j_k) \mapsto (j_1 + 2, j_2, \ldots, j_k)$ injects compositions of m into compositions of $m+2$, and $h_{j_1+2} \geq \eta h_{j_1}$; hence $[t^{m+2}]H^k \geq \eta[t^m]H^k$. The term $k = 0$ contributes only in degree 0. Multiply by g_k and sum. For tanh take $H = \widehat{\phi}_{q_1}$, which satisfies the hypothesis by Lemma 6.3 for odd m and trivially for even m, and let g_k be the coefficients of $\widehat{\phi}_{q_{L-1}} \circ \cdots \circ \widehat{\phi}_{q_2}$ (the identity if $L = 2$). The linear coefficient of a composition of series without constant term is the product of the linear coefficients, and $c_1(q) > 0$ (proof of Corollary 4.2). Iterating from $m = 1$ gives the second bound. □

The constant $\eta(\sigma)$ is crude. Numerically, the smallest ratio $c_{m+2}(\sigma^2)/c_m(\sigma^2)$ over odd $m \leq 4001$ is attained at $m = 1$ and equals 0.0034, 0.0152, 0.0601, 0.1181, 0.1410 for $\sigma = 0.3, 0.5, 1, 2, 3$, against $\eta(\sigma) \approx 1.1\cdot 10^{-4}$, $5.7\cdot 10^{-4}$, $3.8\cdot 10^{-3}$, $1.6\cdot 10^{-2}$, $3.1\cdot 10^{-2}$ (verification record, `verification/revision-2026-10-20/` next to this paper).

Pair masses. For $0 \leq j \leq D$ put $W_j = \binom{D}{j} a_j = \sum_m c_m \mathbb{P}\{\zeta_m = j\}$, where ζ_m is the distance from its start of the walk of Lemma 2.5 after m steps (proof of Lemma 2.6(c)). Then $\sum_j W_j = \sum_m c_m = 1$. If F is odd, $W_j = 0$ for even j, and for odd r the pairing of Lemma 2.5(e) and $\binom{d}{j-1} + \binom{d}{j} = \binom{D}{j}$ give

$$\tau_r = A \sum_{j \geq r+2,\ j \text{ odd}} W_j, \qquad \lambda_{r+1} = \frac{A\, W_{r+2}}{\binom{D}{r+2}}. \tag{12}$$

(If d is even, the unpaired top level $q = d$ fits (12), because $\binom{d}{d} = 1 = \binom{D}{D}$.)

Lemma 6.5 (Cube transfer). *Let F be odd with $c_{m+2} \ge \eta c_m$ for all odd $m \ge 1$.*

(a) For odd j with $j+2 \le D$, $W_{j+2} \ge \eta \frac{(D-j)(D-j-1)}{D^2} W_j$.

(b) For odd r with $r+3 \le D/2$, $\tau_r \le (1+4/\eta)\tau_{r+2}$.

(c) For odd r and an integer $k \ge 0$ with $r+2k+1 \le D/2$, $\tau_r \le (1+4/\eta)^k \tau_{r+2k}$.

Proof. (a) Two outward steps give $\mathbb{P}\{\zeta_{m+2} = j+2\} \ge \mathbb{P}\{\zeta_m = j\}\frac{D-j}{D}\frac{D-j-1}{D}$. Hence $W_{j+2} \ge \sum_{m \text{ odd}} c_{m+2}\mathbb{P}\{\zeta_{m+2} = j+2\} \ge \eta\frac{(D-j)(D-j-1)}{D^2}\sum_m c_m \mathbb{P}\{\zeta_m = j\}$. (b) By (12), $\tau_r = AW_{r+2} + \tau_{r+2}$ and $\tau_{r+2} \ge AW_{r+4}$. By (a), $W_{r+4} \ge \eta\frac{(D-r-2)(D-r-3)}{D^2} W_{r+2} \ge \frac{\eta}{4} W_{r+2}$, since $D-r-3 \ge D/2$. So $AW_{r+2} \le \frac{4}{\eta}\tau_{r+2}$. (c) Apply (b) at $r, r+2, \dots, r+2k-2$. □

Numerically the ratio τ_r/τ_{r+2} stays far below $1+4/\eta$, but it depends strongly on the gains. Over odd r with $r+3 \le D/2$ and $D \in \{41, 101, 201, 301\}$, its largest value is between 1.6 and 1.8 for $L=2$, $\sigma_1 = 8$ (where $1+4/\eta \approx 50$), and between 270 and 320 for $L=2$, $\sigma_1 = 0.2$ (where $1+4/\eta \approx 1.5 \cdot 10^5$); the other configurations in the verification record lie in between.

6.3 The sampled tail Gram matrix

For a nonzero positive semidefinite matrix Σ write $\operatorname{intdim}(\Sigma) = \operatorname{tr}\Sigma/\lambda_{\max}(\Sigma) \ge 1$.

Proposition 6.6 (An intrinsic-dimension expectation bound). *Let $Z_1, \dots, Z_n$ be independent random real symmetric matrices of a common size with $0 \preceq Z_k \preceq \ell I$ almost surely, where $\ell > 0$. Let $Y = \sum_k Z_k$, and let $\Sigma \succeq \mathbb{E}Y$ be positive semidefinite and nonzero. Then for every $\vartheta > 0$,*

$$\mathbb{E}\lambda_{\max}(Y) \;\le\; \frac{e^\vartheta - 1}{\vartheta}\,\lambda_{\max}(\Sigma) + \frac{\ell}{\vartheta}\log \operatorname{intdim}(\Sigma).$$

Proof. We use Lieb's concavity theorem [11, Theorem 6]: for every fixed real symmetric H, the map $B \mapsto \operatorname{tr}\exp(H + \log B)$ is concave on positive definite matrices. Conditionally on $Z_1, \dots, Z_{n-1}$, apply it with $H = \frac{\vartheta}{\ell}\sum_{k<n} Z_k$ and Jensen's inequality to $B = e^{\vartheta Z_n/\ell}$; this replaces $\frac{\vartheta}{\ell}Z_n$ by $\log \mathbb{E}e^{\vartheta Z_n/\ell}$. Iterating over k gives the subadditivity of matrix cumulant generating functions [12, Lemma 3.4],

$$\mathbb{E}\operatorname{tr}\exp\Big(\frac{\vartheta}{\ell}Y\Big) \le \operatorname{tr}\exp\Big(\sum_k \log \mathbb{E}e^{\vartheta Z_k/\ell}\Big).$$

Put $g = (e^\vartheta - 1)/\ell$. Convexity gives $e^{\vartheta x/\ell} \le 1 + gx$ on $[0, \ell]$, so $e^{\vartheta Z_k/\ell} \preceq I + gZ_k$ and $\mathbb{E}e^{\vartheta Z_k/\ell} \preceq I + g\mathbb{E}Z_k$. The logarithm is operator monotone and $\log(1+x) \le x$, so $\log \mathbb{E}e^{\vartheta Z_k/\ell} \preceq g\mathbb{E}Z_k$. As $\operatorname{tr}\exp$ is monotone under $\preceq$, $\mathbb{E}\operatorname{tr}e^{\vartheta Y/\ell} \le \operatorname{tr}e^{g\mathbb{E}Y} \le \operatorname{tr}e^{g\Sigma}$. Since $Y \succeq 0$, $e^{\vartheta\lambda_{\max}(Y)/\ell} - 1 \le \operatorname{tr}(e^{\vartheta Y/\ell} - I)$, and Jensen's inequality gives $e^{\vartheta\mathbb{E}\lambda_{\max}(Y)/\ell} - 1 \le \operatorname{tr}(e^{g\Sigma} - I)$. Let $\mu_{\max} = \lambda_{\max}(\Sigma)$. The function $x \mapsto e^{gx} - 1$ is convex and vanishes at 0, so $e^{g\mu} - 1 \le (\mu/\mu_{\max})(e^{g\mu_{\max}} - 1)$ for every eigenvalue $\mu \in [0, \mu_{\max}]$ of Σ; summing, $\operatorname{tr}(e^{g\Sigma} - I) \le \operatorname{intdim}(\Sigma)(e^{g\mu_{\max}} - 1)$. Since $\operatorname{intdim}(\Sigma) \ge 1$, this gives $e^{\vartheta\mathbb{E}\lambda_{\max}(Y)/\ell} \le \operatorname{intdim}(\Sigma)e^{g\mu_{\max}}$, that is, $\frac{\vartheta}{\ell}\mathbb{E}\lambda_{\max}(Y) \le g\mu_{\max} + \log\operatorname{intdim}(\Sigma)$. Multiply by ℓ/ϑ. □

This is a variant of the expectation bound in the intrinsic-dimension matrix Chernoff inequality [13, Chapter 7]; we prove it to have explicit constants.

Lemma 6.7 (Tail Gram matrix). *Let K be as in Section 3, let $0 \le r < d$, and let $X_1, \dots, X_n$ be independent and uniform on $\mathcal{X}$ (repeats allowed). The sampled matrix $Q_{r,SS} = (Q_r(X_i, X_j))_{i,j\le n}$ of the tail kernel $Q_r = K_{>r}$ satisfies*

$$\mathbb{E}\|Q_{r,SS}\|_{\mathrm{op}} \;\le\; (e-1)\,n\lambda_{r+1} + \tau_r \log\frac{\tau_r}{\lambda_{r+1}}.$$

Proof. Let $\psi(x) = (\lambda_{|U|}^{1/2}\chi_U(x))_{|U|>r}$, so that $Q_r(x,y) = \langle\psi(x),\psi(y)\rangle$. Then $Q_{r,SS} = \Psi\Psi^\mathsf{T}$, where Ψ has rows $\psi(X_i)^\mathsf{T}$, and $\|Q_{r,SS}\|_{\mathrm{op}} = \lambda_{\max}(\Psi^\mathsf{T}\Psi) = \lambda_{\max}(Y)$ with $Y = \sum_i Z_i$ and $Z_i = \psi(X_i)\psi(X_i)^\mathsf{T}$. The Z_i are independent and positive semidefinite, with $\lambda_{\max}(Z_i) = \|\psi(X_i)\|^2 = \tau_r$. Since $\mathbb{E}\chi_U\chi_V = \mathbf{1}_{U=V}$, $\mathbb{E}Y = \Sigma := n\,\mathrm{diag}(\lambda_{|U|})_{|U|>r}$, with $\lambda_{\max}(\Sigma) = n\lambda_{r+1}$ by monotonicity and $\mathrm{intdim}(\Sigma) = \tau_r/\lambda_{r+1}$. Apply Proposition 6.6 with $\ell = \tau_r$ and $\vartheta = 1$. □

The lemma holds for every n and involves no far-pair, near-pair or collision analysis; compare Lemma 7.1. By (15) below, $\|Q_{r,SS}\|_{\mathrm{op}} \geq \max\{n\lambda_{r+1}, \tau_r\}$ for every sample, so the bound is sharp in expectation up to the logarithmic term.

Lemma 6.8 (Intrinsic dimension of the tanh tail)**.** *Let F be odd and satisfy (11), and let r be odd with $r+2 \leq D$. Then*

$$\binom{D}{r+2} \leq \frac{\tau_r}{\lambda_{r+1}} \leq \frac{D^{r+2}}{(r+2)!\,c_{r+2}}, \qquad \textit{hence} \qquad \log\frac{\tau_r}{\lambda_{r+1}} \leq (r+2)\log\frac{eD}{r+2} + \frac{r+1}{2}\log\frac{1}{\eta} + \log\frac{1}{c_1}.$$

Proof. By (12), $\tau_r/\lambda_{r+1} = \binom{D}{r+2}\sum_{j\geq r+2} W_j/W_{r+2}$ (odd j), and $W_{r+2} \leq \sum_{j\geq r+2} W_j \leq \sum_j W_j = 1$. The walk moves away from its start in each of its first $r+2$ steps with probability $\prod_{i=0}^{r+1}\frac{D-i}{D} = (r+2)!\binom{D}{r+2}/D^{r+2}$, so $W_{r+2} \geq c_{r+2}(r+2)!\binom{D}{r+2}/D^{r+2}$. Finally $(r+2)! \geq ((r+2)/e)^{r+2}$, and $c_{r+2} \geq \eta^{(r+1)/2}c_1$ by (11). □

Corollary 4.2 gives instead $\log(1/c_{r+2}) \leq C_\phi\sqrt{r+2} + \log(1/\kappa)$. Either way the leading term is $\log\binom{D}{r+2}$, which the lower bound shows to be sharp; at linear r it is of order d.

6.4 The posterior mean at the training inputs

For a training sequence S and an integer r, let $\mathrm{Coer}(r)$ be the event

$$\frac{1}{n}\sum_{i=1}^{n}(f(X_i)P(X_i))_-^2 \;\geq\; \frac{1}{4}\|P\|_2^2 \qquad \text{for every } P:\mathcal{X}\to\mathbb{R} \text{ with } \deg P \leq r,$$

where $t_- = \max\{-t, 0\}$. By [1, Lemma 4.4], for each $\delta > 0$ there is a constant A_δ, which we may take to be at least 1, such that $\mathbb{P}\{\mathrm{Coer}(r)\} \geq 1-\delta$ whenever $r \leq d/64$, $n \geq A_\delta D_r\log(2D_r)$ and the inputs are independent and uniform. For a sequence S with s distinct inputs $S' = (x'_1,\dots,x'_s)$ and for $0 \leq v < d$, write $Q_v = K_{>v}$ and $Q_{v,SS} = (Q_v(X_i,X_j))_{i,j\leq n}$, repeats included.

Lemma 6.9 (The posterior mean at the training inputs)**.** *Let K be as in Section 3. For every training sequence S of length $n \geq 1$ and every $0 \leq v < d$:*

(a) $m = \sum_{k\leq s} w_k K(\cdot, x'_k)$ *with* $w = K_{S'S'}^{-1}m_{S'}$, *and* $\|m\|_K^2 = w^\mathsf{T}K_{S'S'}w \leq s$;

(b) $\frac{1}{n}\sum_{i=1}^n m_{>v}(X_i)^2 \leq \frac{s}{n}\|Q_{v,SS}\|_{\mathrm{op}}$;

(c) on $\mathrm{Coer}(v)$, $\|m_{\leq v}\|_2^2 \leq 4\frac{s}{n}\|Q_{v,SS}\|_{\mathrm{op}} \leq 4\|Q_{v,SS}\|_{\mathrm{op}}$.

Proof. (a) The conditioning event $\mathcal{A}_S = \{f(X_i)G(X_i) > 0\ \forall i\}$ depends only on $G_{S'}$, so Gaussian regression gives $m(x) = \mathbb{E}[\mathbb{E}[G(x) \mid G_{S'}] \mid \mathcal{A}_S] = K_{xS'}K_{S'S'}^{-1}m_{S'}$. Then $\|m\|_K^2 = w^\mathsf{T}K_{S'\mathcal{X}}K^{-1}K_{\mathcal{X}S'}w = w^\mathsf{T}K_{S'S'}w = m_{S'}^\mathsf{T}K_{S'S'}^{-1}m_{S'}$. The vector $\xi = K_{S'S'}^{-1/2}G_{S'}$ is standard Gaussian, $\mathcal{A}_S$ is a cone in ξ, and the radius of ξ is independent of its direction; so $\mathbb{E}[\|\xi\|^2 \mid \mathcal{A}_S] = s$, and Jensen's inequality gives $m_{S'}^\mathsf{T}K_{S'S'}^{-1}m_{S'} = \|\mathbb{E}[\xi \mid \mathcal{A}_S]\|^2 \leq s$. (b) Projecting (a) onto degrees above v gives $m_{>v} = \sum_k w_k Q_v(\cdot, x'_k)$. Let ν_k be the multiplicity of x'_k in S, and define $\tilde{w} \in \mathbb{R}^n$ by $\tilde{w}_i = w_k/\nu_k$

when $X_i = x'_k$. Then $(Q_{v,SS}\tilde{w})_i = m_{>v}(X_i)$ and $\tilde{w}^{\mathsf{T}} Q_{v,SS}\tilde{w} = w^{\mathsf{T}} Q_{v,S'S'} w \le w^{\mathsf{T}} K_{S'S'} w \le s$, because $K = K_{\le v} + Q_v$ with both terms positive semidefinite. Hence $\sum_i m_{>v}(X_i)^2 = \tilde{w}^{\mathsf{T}} Q_{v,SS}^2 \tilde{w} \le \|Q_{v,SS}\|_{\mathrm{op}}\, \tilde{w}^{\mathsf{T}} Q_{v,SS}\tilde{w} \le s\|Q_{v,SS}\|_{\mathrm{op}}$. (c) Put $P = m_{\le v}$. At every training input $f(X_i)m(X_i) > 0$, because $f(X_i)G(X_i) > 0$ on $\mathcal{A}_S$ and $\mathbb{P}(\mathcal{A}_S) > 0$. Hence $(f(X_i)P(X_i))_- \le |m(X_i) - P(X_i)| = |m_{>v}(X_i)|$: if $fP < 0 < fm$, then $|m - P| = fm - fP > -fP$. Now apply $\mathrm{Coer}(v)$ and (b). □

Proposition 6.10 (Master inequality)**.** *Let K be as in Section 3. Let $0 \le r \le u < d$ with $u \ge 1$ and $1 \le J \le e^2 u$ be integers, and let S have length $n \ge 1$ and lie in $\mathrm{Coer}(r)$. Then*

$$F_\theta(S) \;\ge\; \frac{1}{2} - e\sqrt{\frac{Ju}{\pi d}} - \frac{1}{2}e^{-Jd/(8e^2u)} - \frac{8e^{-J}}{c_\theta^2 \tau_u}\bigl(4\|Q_{r,SS}\|_{\mathrm{op}} + n\lambda_{r+1}\bigr) - \Bigl(2 + \frac{8}{c_\theta^2}\Bigr)\frac{n\lambda_{u+1}}{\tau_u}.$$

Proof. Let $h = m_{\le u}$, $b = \frac{c_\theta}{2}\sqrt{\tau_u/2}$, $E_1 = \{\sigma_x^2 < \tau_u/2\}$ and $E_2 = \{|m - h| \ge b\}$, with σ_x^2 as in Lemma 3.1. By Lemma 3.1(iii), (iv) and Markov's inequality, $\mathbb{P}(E_1) \le 2n\lambda_{u+1}/\tau_u$ and $\mathbb{P}(E_2) \le n\lambda_{u+1}/b^2 = 8n\lambda_{u+1}/(c_\theta^2\tau_u)$. Off $E_1 \cup E_2$, $q_S(x) > \theta$ implies $f(x)m(x) > c_\theta\sigma_x \ge 2b$ by Lemma 3.1(ii), hence $f(x)h(x) > b$. So $\{q_S > \theta\} \subseteq \{fh \ge b\} \cup E_1 \cup E_2$. The function h has degree at most u, and by Lemma 6.9(c) and Lemma 3.1(iv) with $s \le n$,

$$\mathcal{R} = \frac{1}{b^2}\sum_{U \ne \emptyset} \widehat{h}(U)^2 \le \frac{\|m_{\le r}\|_2^2 + \|m_{\le u} - m_{\le r}\|_2^2}{b^2} \le \frac{8}{c_\theta^2\tau_u}\bigl(4\|Q_{r,SS}\|_{\mathrm{op}} + n\lambda_{r+1}\bigr).$$

Corollary 6.2 bounds $\mathbb{P}_x\{fh \ge b\}$, and $F_\theta(S) = 1 - \mathbb{P}_x\{q_S > \theta\}$. □

6.5 The theorem

Let $\phi = \tanh$, put $c_* = 10^{-3}$, $t = \lfloor\sqrt{d}/5\rfloor$, $k_d = \lceil \log d \rceil$,

$$n_0(d) = \frac{\binom{d}{t}}{23C_0}, \qquad n_{\max}(d) = \exp\Bigl(c_*\frac{d\log\log d}{\log d}\Bigr),$$

$$\varepsilon_0 = \frac{1}{64(2 + 8/c_\theta^2)}, \qquad \mathsf{B}_* = 32\bigl((e-1)A_{1/8} + 1\bigr) + A_{1/8},$$

with C_0 as in Theorem 3.2 and $A_{1/8} \ge 1$ the constant of [1, Lemma 4.4] at $\delta = 1/8$. Let $\eta = \eta(\sigma_1)$ and c_1 be as in Lemma 6.4. For odd r put

$$\Lambda_r = 2\log 4 + 2(r+2)\log\frac{ed}{r+2}, \qquad \Lambda'_r = (r+2)\log\frac{eD}{r+2} + \frac{r+1}{2}\log\frac{1}{\eta} + \log\frac{1}{c_1},$$

$$\mathsf{B}_r = 32\bigl((e-1)A_{1/8}\Lambda_r + \Lambda'_r\bigr) + A_{1/8}\Lambda_r,$$

and let

$$\mathsf{T}_d = \log\Bigl(1 + \frac{4}{\eta}\Bigr)\Bigl(1 + \frac{\log d + \log(A_{1/8}/\varepsilon_0)}{2\log\frac{\log d}{1.3c_*}}\Bigr),$$

$$\mathsf{J}_d = \log d + 1 + \log\frac{256\mathsf{B}_*}{c_\theta^2} + \mathsf{T}_d, \qquad \mathsf{r}_d = 1.1\Bigl(\frac{c_*}{\log d} + \frac{2}{d}\Bigr).$$

Let $d_0 \ge 3000$ be such that the following hold for every $d \ge d_0$.

(L1) $A_{1/8}(d+1)\log(2d+2) < n_0(d)$.

(L2) $\frac{4A_{1/8}}{125}\frac{\log d}{d} \le \frac{1}{23C_0}$.

(L3) $c_* \frac{d}{\log d} + 2 \le \min\{\frac{d}{64}, \frac{d+1}{3}\}$ and $\lceil c_* d/\log d \rceil \le d/e$.

(L4) $2k_d \le 0.1(t-2)$.

(L5) $1.1(c_* \frac{d}{\log d} + 2) \le 0.1d$.

(L6) $1.25(\frac{c_*}{\log d} + \frac{2}{d}) \le \frac{1.3c_*}{\log d}$ and $\log \frac{\log d}{1.3c_*} \ge 1$.

(L7) $\Lambda_r \le d$ and $\Lambda'_r \le d$ for every odd $r < c_* d/\log d$.

(L8) $A_{1/8}/\varepsilon_0 \le d$.

(L9) $\mathsf{J}_d \le e^2(t-4)$, $\mathsf{J}_d \mathsf{r}_d \le \pi/(256e^2)$ and $\mathsf{r}_d \le 1/(8e^2 \log 32)$.

Such a d_0 exists. Indeed, $\log n_0(d) \ge \frac{\sqrt{d}}{10} \log d - \frac{1}{2} \log d - \log(23C_0)$ (proof of Corollary 3.4); Λ_r and Λ'_r are $O(c_* d \log\log d/\log d)$ for $r < c_* d/\log d$; $\mathsf{T}_d = (1+o(1))\frac{1}{2}\log(1+\frac{4}{\eta})\frac{\log d}{\log\log d}$, so $\mathsf{J}_d = (1+o(1))\log d$; and $\mathsf{r}_d = (1+o(1))1.1c_*/\log d$. Hence $\mathsf{J}_d \mathsf{r}_d \to 1.1c_* < \pi/(256e^2) = 1.66\ldots \cdot 10^{-3}$, and every other condition holds for large d as well.

Theorem 6.11 (Sign-only learner, typical samples)**.** *Let $\phi = \tanh$, and fix $L \ge 2$ and gains $\sigma_1, \ldots, \sigma_L > 0$. There is d_0, depending only on σ_1, $c_1 = \prod_{\ell<L} c_1(q_\ell)$ and $A_{1/8}$, such that for every $d \ge d_0$ and every integer $0 \le n \le n_{\max}(d) = \exp(10^{-3} d \log\log d/\log d)$ the following holds. If $X_1, \ldots, X_n$ are independent and uniform on $\mathcal{X}$ and $f = \chi_{[d]}$, then*

$$\mathbb{P}_S\Big\{\mathbb{P}_x\{q_S(x) \le \tfrac{13}{20}\} \ge \tfrac{3}{8}\Big\} \;\ge\; \frac{3}{4},$$

and on the same event the sign-only network learner satisfies $\liminf_{N\to\infty} F_N(S) \ge 3/8$. If $n \le n_0(d)$, the inner inequality holds for every training sequence.

Through c_1, the threshold d_0 depends on the depth and on $\sigma_1, \ldots, \sigma_{L-1}$, but not on σ_L; indeed q_S does not change when K is multiplied by a constant. The limit $N \to \infty$ is taken at fixed d, L, gains and S. By Remark 3.5, the conclusion $\liminf_N F_N(S) \ge 3/8$ also holds, on the same event and with the same status, for the margin learner whose readout variance is $\sigma_L^2 N^{-b}$ with $b < 1$: its correct-label probabilities converge to the same $q_S(x)$. Two lemmas come first.

Lemma 6.12 (Adaptive cutoff)**.** *Let $d \ge d_0$ and $n_0(d) < n \le n_{\max}(d)$.*

(a) Some odd $r \ge 1$ has $A_{1/8} D_r \log(2D_r) \le n$; let r be the largest one.

(b) $t - 4 \le r < c_* d/\log d$; in particular $r \le d/64$ and $r + 2 \le (d+1)/3$.

(c) $n/\binom{D}{r+2} < A_{1/8}\Lambda_r$ and $n\lambda_{r+1}/\tau_r < A_{1/8}\Lambda_r$.

(d) Put $\hat\rho = 1.25(r+2)/d$ and $k^* = \lceil \log(A_{1/8}\Lambda_r/\varepsilon_0)/(2\log(1/\hat\rho)) \rceil$. Then $0 \le k^* \le k_d$, and $u = r + 2k^*$ satisfies $u+1 \le D/2$, $\rho_u = u/d \le 1.1(r+2)/d \le \mathsf{r}_d$ and $n\lambda_{u+1}/\tau_u \le n/\binom{D}{u+2} < \varepsilon_0$.

Proof. Two binomial facts are used. First, $\binom{d}{q-1} = \frac{q}{d-q+1}\binom{d}{q} \le \frac{1}{2}\binom{d}{q}$ for $1 \le q \le (d+1)/3$, hence $D_v \le 2\binom{d}{v}$ for $v \le (d+1)/3$. Second, $\binom{D}{j+2}/\binom{D}{j} = \frac{(D-j)(D-j-1)}{(j+1)(j+2)} \ge (\frac{D-j-1}{j+2})^2$.

(a) $D_1 = d+1$, and $A_{1/8} D_1 \log(2D_1) < n_0(d) < n$ by (L1).

(b) Suppose $r \ge r_1 := \lceil c_* d/\log d \rceil$. Since $A_{1/8} \ge 1$ and $\log(2D_r) \ge 1$, $n \ge D_r \ge \binom{d}{r_1} \ge (d/r_1)^{r_1}$. The function $x \mapsto x\log(d/x)$ increases on $(0, d/e]$, and $c_* d/\log d \le r_1 \le d/e$ by (L3),

so $\log n \ge \frac{c_* d}{\log d}\log\frac{\log d}{c_*} > \log n_{\max}(d)$, a contradiction. Hence $r < c_* d/\log d$, and (L3) gives the two consequences. Now suppose $r+2 \le t' := t-3$. Then $t' \le (d+1)/3$, so $D_{t'} \le 2\binom{d}{t'}$ and $\log(2D_{t'}) \le \log 4 + t'\log(ed) \le \frac{\sqrt{d}}{4}\log d$ for $d \ge 3000$. Also $\binom{d}{t'} \le \binom{d}{t}(\frac{t}{d-t+1})^3 \le \binom{d}{t}\frac{8}{125d^{3/2}}$. By maximality of r and (L2),

$$n < A_{1/8} D_{r+2}\log(2D_{r+2}) \le A_{1/8} D_{t'}\log(2D_{t'}) \le \frac{4A_{1/8}}{125}\frac{\log d}{d}\binom{d}{t} \le n_0(d),$$

a contradiction. Hence $r \ge t-4$.

(c) By maximality $n < A_{1/8}D_{r+2}\log(2D_{r+2})$, while $D_{r+2} \le 2\binom{d}{r+2} \le 2\binom{D}{r+2}$ and $\log(2D_{r+2}) \le \log 4 + (r+2)\log\frac{ed}{r+2}$ by $\binom{d}{v} \le (ed/v)^v$. The second claim follows from Lemma 2.6(b).

(d) By (L6), $\hat\rho \le 1.3c_*/\log d$ and $\log(1/\hat\rho) \ge 1$; by (L7) and (L8), $\log(A_{1/8}\Lambda_r/\varepsilon_0) \le 2\log d$. So $k^* \le k_d$, and $k^* \ge 0$ because $A_{1/8}\Lambda_r > \varepsilon_0$. For odd $j \in [r+2, u]$, (L4) and $r \ge t-4$ give $j+2 \le r+2k_d+2 \le 1.1(r+2)$, and (L5) gives $D-j-1 \ge d-1.1(r+2) \ge 0.9d$. Hence $(j+2)/(D-j-1) \le \frac{1.1}{0.9}\frac{r+2}{d} < \hat\rho$ and $\binom{D}{j+2} \ge \hat\rho^{-2}\binom{D}{j}$. Iterating, $\binom{D}{u+2} \ge \hat\rho^{-2k^*}\binom{D}{r+2}$, so by (c) $n/\binom{D}{u+2} < A_{1/8}\Lambda_r\hat\rho^{2k^*} \le \varepsilon_0$ by the choice of k^*. Lemma 2.6(b) at u gives $n\lambda_{u+1}/\tau_u \le n/\binom{D}{u+2}$. Finally $u+1 < u+2 \le 1.1(r+2) \le D/2$, and $\rho_u \le 1.1(r+2)/d \le \mathsf{r}_d$ by (b). □

Lemma 6.13 (Regularity across the band)**.** *In the setting of Lemma 6.12,* $\log(\tau_r/\tau_u) \le k^*\log(1+4/\eta) \le \mathsf{T}_d$.

Proof. Lemma 6.5(c) applies, with the η of (11), because $u+1 \le D/2$. By definition $k^* \le 1 + \log(A_{1/8}\Lambda_r/\varepsilon_0)/(2\log(1/\hat\rho))$; use $\Lambda_r \le d$ (L7) and $\hat\rho \le 1.3c_*/\log d$ (L6). □

So $\tau_r/\tau_u \le e^{\mathsf{T}_d} = d^{o(1)}$, uniformly in n, with $u - r = 2k^* = O(\log d/\log(1/\rho))$ for $\rho = r/d$. The convergence is slow: at $\sigma_1 = 1$, $A_{1/8} = 1$ and $d = 10^{58}$, $\mathsf{T}_d = 51.0$, while the leading term $\frac{1}{2}\log(1+\frac{4}{\eta})\log d/\log\log d$ is 94.9. The difference comes mainly from the term $\log(1/(1.3c_*)) \approx 6.6$ in the denominator $2\log(\log d/(1.3c_*))$ of T_d, which at this d still exceeds $\log\log d \approx 4.9$.

Proof of Theorem 6.11. The statement about $\liminf_N F_N(S)$ follows from Proposition 2.4(iii), since $\{q_S \le 13/20\} \subseteq \{q_S < 2/3\}$.

Regime I: $n \le n_0(d)$. Apply Theorem 3.2 with $r = t-1$; here $t \ge 1$ because $d \ge 25$. Lemma 3.3 gives $\mathrm{err}_{t-1}(\chi_{[d]}) \ge \frac{1}{2} - \frac{t}{\sqrt{2\pi d}} \ge \frac{1}{2} - \frac{1}{5\sqrt{2\pi}} > 0.4202$, and $C_0 n/\binom{d}{t} \le 1/23 < 0.0435$. So $F_\theta(S) > 0.3767 > 3/8$ for every training sequence.

Regime II: $n_0(d) < n \le n_{\max}(d)$. Take r, k^* and u from Lemma 6.12, and let $\mathsf{E}_{\mathrm{coer}} = \mathrm{Coer}(r)$ and $\mathsf{E}_{\mathrm{gram}} = \{\|Q_{r,SS}\|_{\mathrm{op}} \le 8\,\mathbb{E}\|Q_{r,SS}\|_{\mathrm{op}}\}$. Since $r \le d/64$ and $n \ge A_{1/8}D_r\log(2D_r)$, [1, Lemma 4.4] gives $\mathbb{P}(\mathsf{E}_{\mathrm{coer}}) \ge 7/8$, and Markov's inequality gives $\mathbb{P}(\mathsf{E}_{\mathrm{gram}}) \ge 7/8$. Fix $S \in \mathsf{E}_{\mathrm{coer}} \cap \mathsf{E}_{\mathrm{gram}}$, an event of probability at least $3/4$.

Step 1 (head and band). By Lemma 6.7, Lemma 6.8 (which gives $\log(\tau_r/\lambda_{r+1}) \le \Lambda'_r$) and Lemma 6.12(c),

$$4\|Q_{r,SS}\|_{\mathrm{op}} + n\lambda_{r+1} \le 32\Big[(e-1)n\lambda_{r+1} + \tau_r\log\frac{\tau_r}{\lambda_{r+1}}\Big] + n\lambda_{r+1} \le \tau_r\mathsf{B}_r \le \mathsf{B}_* d\,\tau_r,$$

the last step by (L7).

Step 2 (amplitude). By Lemma 6.13,

$$\frac{8}{c_\theta^2\tau_u}(4\|Q_{r,SS}\|_{\mathrm{op}} + n\lambda_{r+1}) \le \bar{\mathcal{R}} := \frac{8\mathsf{B}_r}{c_\theta^2}\Big(1+\frac{4}{\eta}\Big)^{k^*}.$$

Let $J = \max\{1, \lceil \log(32\bar{\mathcal{R}}) \rceil\}$. Then $\bar{\mathcal{R}} e^{-J} \le 1/32$. Since $\mathsf{B}_r \ge \Lambda_r \ge 2 \log 4$ and $c_\theta < 1$, we have $32\bar{\mathcal{R}} \ge 256 \mathsf{B}_r / c_\theta^2 > 1$, so $J = \lceil \log(32\bar{\mathcal{R}}) \rceil$, and by Step 1 and Lemma 6.13, $J \le 1 + \log(32\bar{\mathcal{R}}) \le \mathsf{J}_d$.

Step 3 (admissibility of J). By (L9) and Lemma 6.12(b), $J \le \mathsf{J}_d \le e^2(t-4) \le e^2 r \le e^2 u$.

Step 4. Apply Proposition 6.10 with these r, u and J. By (L9) and Lemma 6.12(d), $e\sqrt{Ju/(\pi d)} \le e\sqrt{\mathsf{J}_d \mathsf{r}_d / \pi} \le 1/16$ and $\frac{1}{2} e^{-Jd/(8e^2 u)} \le \frac{1}{2} e^{-1/(8e^2 \mathsf{r}_d)} \le 1/64$. The amplitude term is at most $\bar{\mathcal{R}} e^{-J} \le 1/32$, and the last term is at most $(2 + 8/c_\theta^2)\varepsilon_0 = 1/64$. Hence $F_\theta(S) \ge \frac{1}{2} - \frac{1}{16} - \frac{1}{64} - \frac{1}{32} - \frac{1}{64} = \frac{3}{8}$. □

Remark 6.14 (Status, and the role of the coercivity lemma). Regime I uses only Lemma 3.3 and Theorem 3.2, and is unconditional. Regime II uses [1, Lemma 4.4], whose proof runs through [1, Lemma 4.3] to [1, Lemma 4.1]. That lemma does not apply a stated theorem of [10]: it uses the intermediate metric estimate (2.20) in the proof of [10, Lemma 2.12], with the energy normalization of that proof undone, and [1, Remark 4.2] justifies this homogeneous form. We have not re-checked it against [10]. Theorem 6.11 is therefore proved conditionally on [1, Lemma 4.1]; the subcritical theorem [1, Theorem 1.1] rests on the same lemma, and so do the lower bounds on the conditioning cost in Section 7 (Propositions 7.2 and 7.3 and Corollaries 7.4 and 7.5). The dependence is not cosmetic. A contraction argument gives squared coercivity only for $n \gtrsim D_r^2$. With that sample size the band term $n\lambda_{r+1}/\tau_r$ would be of order $D_r^2 / \binom{D}{r+2}$, the amplitude ratio would have logarithm of order $r \log(d/r)$, Corollary 6.2 would force $r^2 \log(d/r) \lesssim d$, and the argument would reach only $\log n = O(\sqrt{d \log d})$, inside the range of Theorem 3.2. (This counts the method's own bounds; it is not a statement about the learner.)

Remark 6.15 (Constants, and the size of d_0). (1) The constant $c_* = 10^{-3}$ is not optimized. The budget constants $1/32$, $1/64$ and ε_0 can be made arbitrarily small at the cost of additive constants in J, and the factor 1.1 in r_d can be brought to $1 + o(1)$, so the same proof gives every fixed $c_* < \pi/(64e^2) = 6.6\ldots \cdot 10^{-3}$. The method stops there because $\log \bar{\mathcal{R}} = (1 + o(1)) \log d$, so Corollary 6.2 forces $\rho_u \lesssim 1/\log d$. (2) The threshold d_0 is not explicit, because $A_{1/8}$ is not, and the conditions (L1)–(L9) make it astronomically large even with the smallest value the proof allows, $A_{1/8} = 1$. For $L = 2$, the condition $\mathsf{J}_d \mathsf{r}_d \le \pi/(256e^2)$ of (L9) first holds at $\log_{10} d \approx 30.7$, 37.1, 42.3, 58.2, 94.3, 157.1, 273.5 for $\sigma_1 = 8, 3, 2, 1, 0.5, 0.3, 0.2$ when $A_{1/8} = 1$, and at 36.9, 44.3, 50.2, 68.2, 108.7, 177.8, 302.9 when $A_{1/8} = 100$. For these gains and $1 \le A_{1/8} \le 100$ all other conditions hold once $d \ge 2.6 \cdot 10^7$. The cause is the slow convergence of $\mathsf{J}_d / \log d$ to 1: at $\sigma_1 = 1$, $A_{1/8} = 1$ and $d = 10^{58}$, $\mathsf{J}_d / \log d = 1.511$, against the allowed value 1.510. (3) Even apart from d_0, the theorem adds to Theorem 3.2 only where $n_{\max}(d) > n_0(d)$, that is, for d above a crossover $d_\times \approx 2.58 \cdot 10^8$; the ratio $\log n_{\max}(d) / \log n_0(d)$ is 0.67, 1.78 and 4.82 at $d = 10^8, 10^9, 10^{10}$. A direct evaluation of the inequalities that Regime II actually uses, with the actual r, k^*, u and J in place of the uniform bounds (L4)–(L9), found all of them satisfied at 26 grid dimensions between $10^{8.42}$ and 10^{180}, for $L = 2$, $\sigma_1 \in \{0.3, 1, 3\}$ and $A_{1/8} \in \{1, 100\}$; the largest value of Ju/d was $6.6 \cdot 10^{-4}$, against the requirement $1.66 \cdot 10^{-3}$. This is a numerical observation at grid points, conditional on the unknown value of $A_{1/8}$; it is not part of the theorem. The scripts are in the verification record (`verification/revision-2026-10-20/scripts/fix_S1`).

Remark 6.16 (Why the restriction lemma succeeds where clipped parity does not). For h of degree at most u, the clipped-parity lemma [1, Lemma 6.1] gives $\mathbb{P}_x\{fh \ge b\} \le \frac{1}{2}(1 + \mathbb{E} f \operatorname{clip}(h/b)) \le \frac{1}{2} + \frac{\rho_u}{2} \mathcal{R}^{1/2}$, with $\mathcal{R}$ as in Corollary 6.2; the second step uses the Laplacian form [1, eq. (23)], which does not see the constant term. This bound is linear in the amplitude $\mathcal{R}^{1/2}$, and Corollary 6.2 is logarithmic in it. For the arcsine kernel the sharp ratio $\theta_r = \binom{D}{r+2} \lambda_{r+1}/\tau_r \asymp 1/r$ [1, Lemma 3.2] cancels the factor r of the coercivity threshold, the amplitude ratio is bounded at fixed $\rho = r/d$, and the linear bound suffices at linear degree [1, Theorem 10.2]. For tanh the bounds proved here give only an amplitude ratio of order $\mathsf{B}_r \, d^{o(1)}$, where B_r is of order rH_r, with $H_r = \log(eD/r)$, through

both $n/\binom{D}{r+2}$ (the logarithm in the coercivity threshold) and $\log(\tau_r/\lambda_{r+1}) \geq \log\binom{D}{r+2}$ (the intrinsic dimension). With the linear bound one would need $\rho_u(rH_r)^{1/2}d^{o(1)}$ to be small, which confines r to at most $d^{2/3+o(1)}$. With the logarithmic bound one needs only $\rho_u \log d$ small, which allows r up to $c\,d/\log d$; this is the source of the exponent $d\log\log d/\log d$.

Remark 6.17 (Other activations). The proof uses tanh only through three properties: oddness and Assumption 2.1 (for the pairing and the width limit); two-step regularity of the first-layer coefficients, $c_{m+2}(q_1) \geq \eta c_m(q_1)$ for odd m with some $\eta > 0$; and $c_1(q_\ell) > 0$ for every layer. These give (11), hence $c_m > 0$ for every odd m and $K \succ 0$. No lower bound on individual eigenvalues and no far-pair estimate is used. So Theorem 6.11 holds verbatim, with the same status, for every activation with these properties, with d_0 depending on η and c_1. An example is erf. Its normalized dual is $\widehat{\phi}_q(\rho) = \arcsin(\varsigma_q\rho)/\arcsin\varsigma_q$ with $\varsigma_q = 2q/(1+2q)$, so $c_{2k+3}(q)/c_{2k+1}(q) = \varsigma_q^2(2k+1)^2/((2k+2)(2k+3)) \geq \varsigma_q^2/6$ and $c_1(q) = \varsigma_q/\arcsin\varsigma_q > 0$. The margin learner for erf remains uncovered (Remark 5.6).

Remark 6.18 (What would give $\exp(\Omega(d))$ for the sign-only learner). The following is an *extension suggested by the proof*, conditional on two open estimates; it is not proved. In the proof of Theorem 6.11 the amplitude ratio is polynomial in d for two reasons, the logarithm in the coercivity threshold and the intrinsic dimension (Remark 6.16). Fix $c_\flat \in (0, c_*)$ and consider:

(C) *Log-free one-sided coercivity at linear degree.* There are $\rho_0 \in (0, 1/64]$ and, for each $\delta \in (0,1)$, functions $\varkappa_\delta, c_\delta^{\mathrm{coer}} : (0,\rho_0] \to (0,\infty)$ with $\rho\log(\varkappa_\delta(\rho)/c_\delta^{\mathrm{coer}}(\rho)) \to 0$ as $\rho \to 0$, such that for all $d \geq d_{\mathrm{C}}(\delta,\rho)$, all odd r with $c_\flat d/\log d \leq r \leq \rho d$, and all n with $\varkappa_\delta(\rho)D_r \leq n \leq \varkappa_\delta(\rho)D_{r+2}$, independent uniform inputs satisfy, with probability at least $1-\delta$, $\frac{1}{n}\sum_i (f(X_i)P(X_i))_-^2 \geq c_\delta^{\mathrm{coer}}(\rho)\|P\|_2^2$ for every P of degree at most r.

(G) *Log-free tail Gram bound.* In the same range, with probability at least $1-\delta$, $\|Q_{r,SS}\|_{\mathrm{op}} \leq C_\delta^{\mathrm{Gram}}(\tau_r + n\lambda_{r+1})$.

[1, Lemma 4.4] has the form of (C) with $\varkappa_\delta = A_\delta\log(2D_r)$ and $c_\delta^{\mathrm{coer}} = 1/4$; this factor depends on d, which (C) does not allow. Equation (15) shows that (G) would be sharp up to the constant. Run the proof of Theorem 6.11 with the cutoff $\varkappa_\delta(\rho_*)D_r \leq n < \varkappa_\delta(\rho_*)D_{r+2}$ for a small fixed ρ_*, and with coercivity constant $c_\delta^{\mathrm{coer}}(\rho_*)$ in place of $1/4$ in Lemma 6.9(c). Since $c_\flat < c_*$, for large d every cutoff r that this rule assigns to an $n > n_{\max}(d)$ satisfies $r \geq c_\flat d/\log d$, the lower limit in (C), so the sample sizes covered by (C) join those of Theorem 6.11 (a lower limit $d/\log d$ in (C) would leave a gap between the two ranges). Then $n/\binom{D}{r+2} \leq 2\varkappa_\delta(\rho_*)$; (G) bounds the head; a band of bounded width $u - r = O(\log(\varkappa_\delta(\rho_*)/\varepsilon_0)/\log(1/\rho_*))$ makes the last term of Proposition 6.10 small; and Lemma 6.5(c) bounds τ_r/τ_u independently of d. The amplitude ratio is then bounded independently of d, and Corollary 6.2 closes for $r \leq \rho_* d$ once ρ_* is small. This would give $\mathbb{P}_S\{F_\theta(S) \geq 3/8\} \geq 3/4$ for all $n \leq e^{cd}$, an $\exp(\Omega(d))$ bound. In the exploration record of this revision (folder `verification/revision-2026-10-20/`: `exploration-synthesis.md`, Section 2.2, with the estimate stated in `lemma-T-draft.md`), (G) is replaced by a sharp level-mass estimate, $\theta_r \leq C/\sqrt{r}$ together with $W_i \leq CW_{r+2}$ for odd $i \in [r+2, D/2]$, combined with the Tropp-type bound behind Lemma 7.1; that variant is sketched there. We do not prove the level-mass estimate (`lemma-T-draft.md` is an unrefereed draft argument), and the computations of Section 8 do not settle its upper bound $\theta_r \leq C/\sqrt{r}$: they show $\theta_r \geq c/\sqrt{r}$ in the computed range, but $\theta_r\sqrt{r}$ still grows slowly at its upper end. Hypothesis (C) alone does not suffice with the Gram bounds proved here: Lemma 6.7 retains the term $\tau_r\log(\tau_r/\lambda_{r+1}) \geq \tau_r\log\binom{D}{r+2}$, of order $d\,\tau_r$ at linear degree, and the bound behind Lemma 7.1 retains a term polynomial in d unless the level masses are regular in the above sense.

7 Where the exponential proof breaks

The $\gamma = 0$ part of the subcritical theorem [1, Theorem 9.2] covers n up to e^{cd}. It chooses an adaptive odd cutoff r with $n \approx A_\delta D_r \log D_r$, where $D_r = \sum_{q \le r} \binom{d}{q}$. Its key normalized estimate is

$$\frac{\mathcal{D}(S)}{n} \le C d \rho^{-\beta}, \qquad \rho = r/d, \qquad \text{[1, eq. (42)]},$$

a bound polynomial in d (at most $Cd^{1+\beta}$). It combines this with the head bound $\mathbb{E}^* \|G_{\le r}\|_2^2 \le 8\tau_r + 16 b_r \tau_r \mathcal{D}/n$ ([1, eq. (45)] states it with that paper's Gram bound [1, eq. (44)]; the form with $b_r\tau_r$ follows in the same way from [1, eqs. (41) and (43)]), where $\mathbb{E}^*$ is the posterior expectation and $b_r\tau_r$ bounds the sampled Gram matrix of the tail kernel. The clipped parity lemma is then applied at the scale of the posterior output amplitude, which is of order $d^{(1-\beta)/2}$ there [1, Lemma 7.5]. That amplitude step uses the variance $\tau_u \asymp d^{-\beta}$ of the Fourier levels above $u \approx d/50$; at the standard scaling this variance is tiny, and [1, Lemma 7.5] gives no useful amplitude. We show below that the normalization [1, eq. (42)] also fails: the conditioning cost per sample is superpolynomial in d. We then prove a sharper lower bound, of order a^2/τ_r per sample up to a factor $\min\{1, A_\delta \sqrt{r H_r/d}\}$, with $H_r = \log(eD/r)$, that is of order one when $\log n \asymp d$. So no upper bound on the cost can match the first lower bound, and with our tail-Gram bound the subcritical head estimate at the margin scale cannot give an exponential bound (it gives nothing at linear cutoffs), whatever upper bound on the cost holds. Like Theorem 6.11 for $n > n_0(d)$, the lower bounds on the cost in Propositions 7.2 and 7.3 and Corollaries 7.4 and 7.5 use the squared coercivity lemma [1, Lemma 4.4], so they rest on [1, Lemma 4.1] (Remark 6.14). Lemma 7.1, the upper bound in Proposition 7.2 and (15) do not use it.

Fix $\delta \in (0, 1/8)$, let $A_\delta \ge 1$ be the sampling constant of [1, Lemma 4.4] (squared coercivity: with probability at least $1-\delta$, $\frac{1}{n}\sum_i (f(X_i)P(X_i))_-^2 \ge \frac{1}{4}\|P\|_2^2$ for all P of degree at most r, if $r \le d/64$ and $n \ge A_\delta D_r \log(2D_r)$), and let $r = r(n)$ be the largest odd integer with $A_\delta D_r \log(2D_r) \le n$.

Lemma 7.1 (Universal tail-Gram bound)**.** *Let F be odd. There are $c_{\mathrm{G}} > 0$ and C_δ, depending only on δ and A_δ, such that for large d, every n with $\exp((\log(d+1))^3) \le n \le e^{c_{\mathrm{G}} d}$, and $r = r(n)$, the following holds. With probability at least $1-\delta$ over iid uniform inputs, the matrix $Q_{SS} = ((K_{>r})(X_i, X_j))_{i,j \le n}$ satisfies*

$$\|Q_{SS}\|_{\mathrm{op}} \le b_r \tau_r, \qquad b_r = C_\delta(1 + \sqrt{d\, r H_r} + r H_r), \qquad H_r = \log(eD/r),$$

and $r \le d/64$, $r \ge (\log d)^2/4$.

Proof. The range of r follows exactly as in [1, Lemma 6.4], which uses only binomial estimates. That lemma's argument also gives $n < A_\delta D_{r+2} \log(2D_{r+2}) \le C A_\delta \binom{D}{r+2} r H_r$. [1, Lemma 5.2] bounds $\mathbb{P}\{\|Q_{SS} - \tau I\| > z\tau\}$ in terms of four quantities. Its proof uses only these properties of Q: positive semidefinite, constant diagonal $\tau = \tau_r$, $\max_x \sum_y Q(x,y)^2 = M s_{2,r}$, $\|Q\|_{\mathrm{op}} = M\lambda_{r+1}$, and the far-pair bound $\varepsilon_r^{\mathrm{far}} = \max_{d/4 \le \mathrm{dist} \le 3d/4} |Q(x,y)|$. Here

$$\frac{n\lambda_{r+1}}{\tau_r} \le \frac{n}{\binom{D}{r+2}} \le C A_\delta r H_r, \qquad \frac{\sqrt{n \cdot 2\log M \cdot s_{2,r}}}{\tau_r} \le \sqrt{2d \log 2 \, \frac{n}{\binom{D}{r+2}}} \le C\sqrt{A_\delta d\, r H_r},$$

by Lemma 2.6(b),(d). For the far pairs, the proof of [1, Lemma 5.1] gives $\varepsilon_r^{\mathrm{far}} \le \tau_r(C\sqrt{D}e^{-r/8} + e^{-c_0 D})$, once two of its steps are reread. First, its bound for $r < j \le D - r$ uses only the Krawtchouk estimate (20) of that paper together with $\sum_{j>r} \binom{D}{j} a_j = \tau_r/A$ (here $Q = \sum_{j \ge r+2,\ j \text{ odd}} A a_j \mathsf{K}_j$, because F is odd and r is odd). Second, for $j > D - r$ it uses monotonicity: $a_j \le a_b \le \tau_r/(A\binom{D}{b})$ for

an odd b near $D/4$. Thus $2\log M \cdot \varepsilon_r^{\mathrm{far}}/\tau_r \to 0$ for $r \geq (\log d)^2/4$. The near-pair and collision terms are at most $3ne^{d\mathsf{h}(1/4)}/2^d$ and $n^2 \cdot 3e^{d\mathsf{h}(1/4)}/2^d$. They vanish when $2c_{\mathrm{G}} < \log 2 - \mathsf{h}(1/4)$. Choosing z as a δ-dependent multiple of the sum of the four terms gives the claim. □

Proposition 7.2 (The conditioning cost per sample is superpolynomial)**.** *Let $\phi = \tanh$; fix L, gains, a and $\delta \in (0, 1/8)$. For large d and every n with $\exp((\log(d+1))^3) \leq n \leq e^{c_{\mathrm{G}} d}$, with probability at least $1 - 2\delta$ over iid uniform inputs labelled by full parity,*

$$\frac{\mathcal{D}(S)}{n} \;\geq\; \frac{a^2 - 10\tau_r}{\tau_r(16b_r + 4CA_\delta r H_r)}, \qquad r = r(n).$$

Consequently, for every $\epsilon > 0$ and k, for all large d and every n with $\exp(d^\epsilon) \leq n \leq e^{c_{\mathrm{G}} d}$, on the same event, $\mathcal{D}(S)/n \geq d^k$. In the other direction, on the event of Lemma 5.1, $\mathcal{D}(S)/n \leq e^{C\sqrt{d}}$ for every $n \geq 1$ with $n(0.2)^{0.1d} \leq 0.04$. (Some upper limit on n is necessary in the first statement: for fixed d, $\mathcal{D}(S)$ is bounded by the cost of the full sign pattern of f, so $\mathcal{D}(S)/n \to 0$ as $n \to \infty$.)

Proof. Work on the intersection of the squared-coercivity event and the event of Lemma 7.1. Write $p = G_{\leq r}$ and $\mathbb{E}^*$ for the posterior expectation. Every posterior draw has $|G| > a$ everywhere, so $\mathbb{E}^*\|G\|_2^2 \geq a^2$. At the training inputs, $f(X_i)G(X_i) > 0$ implies $(f(X_i)p(X_i))_- \leq |G_{>r}(X_i)|$. Coercivity gives $\|p\|_2^2 \leq \frac{4}{n}\sum_i G_{>r}(X_i)^2$. Lemma 5.3, applied to $n^{-1/2}(G_{>r}(X_i))_i$, whose covariance Q_{SS}/n has trace τ_r and largest eigenvalue at most $b_r\tau_r/n$, gives $\mathbb{E}^*\frac{1}{n}\sum_i G_{>r}(X_i)^2 \leq 2\tau_r + 4b_r\tau_r\mathcal{D}/n$. Applied to $M^{-1/2}G_{>r}$ it gives $\mathbb{E}^*\|G_{>r}\|_2^2 \leq 2\tau_r + 4\lambda_{r+1}\mathcal{D}$, with $\lambda_{r+1} \leq \tau_r/\binom{D}{r+2} \leq \tau_r CA_\delta r H_r/n$. Hence

$$a^2 \leq \mathbb{E}^*\|p\|_2^2 + \mathbb{E}^*\|G_{>r}\|_2^2 \leq 10\tau_r + \frac{\tau_r\mathcal{D}}{n}(16b_r + 4CA_\delta r H_r),$$

which is the first claim. For the second, the tanh dual activations have all moments finite: if $h(g) = \tanh(\sqrt{q}g)$, the identity $\mathbb{E}[h^{(k)}(G)He_j(G)] = \mathbb{E}[h(G)He_{j+k}(G)]$ gives $\sum_m \frac{m!}{(m-k)!}\beta_m^2 = \mathbb{E}[h^{(k)}(G)^2] < \infty$. Regard (c_m) as a probability distribution on degrees. For a composition $\widehat{\phi} \circ H$, the degree distribution is the law of a random sum $\sum_{i \leq Y} J_i$, where Y has the distribution of the coefficients of $\widehat{\phi}$ and the J_i are independent copies of the degree under H, independent of Y. Since $(\sum_{i\leq Y} J_i)^k \leq Y^{k-1}\sum_{i\leq Y} J_i^k$, we get $\mathbb{E}(\sum_{i\leq Y} J_i)^k \leq \mathbb{E}Y^k\,\mathbb{E}J^k$. So $T(r) \leq C_k r^{-k}$ for every k, and $\tau_r \leq AT(r)$. If $n \geq \exp(d^\epsilon)$, then $D_{r+2}\log(2D_{r+2}) > n/A_\delta$ forces $r \geq d^{\epsilon/2}$ for large d, so $\tau_r \leq d^{-k'}$ for every k', while $b_r + rH_r \leq Cd\log d$. The upper bound follows from Lemma 5.2 and $E_b \leq e^{C\sqrt{d}}$. □

Proposition 7.3 (A sharper lower bound on the cost per sample)**.** *Let $\phi = \tanh$; fix L, gains, a and $\delta \in (0, 1/8)$, and let c_{G}, C_δ and b_r be as in Lemma 7.1. For large d and every n with $\exp((\log(d+1))^3) \leq n \leq e^{c_{\mathrm{G}} d}$, with probability at least $1 - 4\delta$ over iid uniform inputs labelled by full parity,*

$$\mathcal{D}(S) \;\geq\; \frac{\delta n a^2}{\tau_r}\min\Big\{\frac{1}{16}, \frac{A_\delta \log(2D_r)}{128\, b_r}\Big\} - \log 2, \qquad r = r(n). \tag{13}$$

Consequently, on the same event,

$$\frac{\mathcal{D}(S)}{n} \;\geq\; c_\delta \frac{a^2}{\tau_r}\min\Big\{1, A_\delta\sqrt{rH_r/d}\Big\}, \qquad c_\delta = \frac{\delta}{2}\min\Big\{\frac{1}{16}, \frac{1}{480C_\delta}\Big\}. \tag{14}$$

Only the training margins $y_iG(X_i) > a$ are used. Moreover, for every sequence S of n inputs and every $0 \leq r < d$,

$$\|Q_{SS}\|_{\mathrm{op}} \;\geq\; \max\{n\lambda_{r+1}, \tau_r\}. \tag{15}$$

Proof. Write $y_i = f(X_i)$, $p = G_{\le r}$, $t = G_{>r}$, and $t_S = (t(X_i))_{i\le n}$, a centered Gaussian vector with covariance Q_{SS}. Put

$$\varepsilon^2 = \sum_{|U|\le r} \Big(\frac{1}{n}\sum_{i=1}^{n} y_i \chi_U(X_i)\Big)^2.$$

Since $y_i\chi_U(X_i) = \chi_{[d]\setminus U}(X_i)$ and $[d]\setminus U \neq \emptyset$, each inner average has mean zero and variance $1/n$, so $\mathbb{E}\varepsilon^2 = D_r/n$. For $i \neq j$ only $U = [d]$ survives in $\mathbb{E}[y_i y_j Q(X_i, X_j)]$, which is therefore λ_d; and $Q(X_i, X_i) = \tau_r$. Hence $\mathbb{E}\, y^{\mathsf T} Q_{SS} y = n\tau_r + n(n-1)\lambda_d \le 2n\tau_r$, because $\tau_r \ge \lambda_d \sum_{q>r} \binom{d}{q} \ge 2^{d-1}\lambda_d$ by monotonicity ($r < d/2$) and $n \le e^{c_{\mathrm G} d} \le 2^{d-1}$ (the proof of Lemma 7.1 takes $2c_{\mathrm G} < \log 2 - \mathsf h(1/4)$, so $c_{\mathrm G} < \frac12 \log 2$). Let E be the intersection of the coercivity event of [1, Lemma 4.4] at degree r (it applies since $r \le d/64$ by Lemma 7.1 and $n \ge A_\delta D_r \log(2D_r)$ by the choice of r), the event of Lemma 7.1, $\{\varepsilon^2 \le D_r/(n\delta)\}$ and $\{y^{\mathsf T} Q_{SS} y \le 2n\tau_r/\delta\}$. By Markov's inequality, $\mathbb{P}(E) \ge 1 - 4\delta$. Fix a sample in E.

Every $G \in A_S$ has $y_i G(X_i) > a$ for all i. Cauchy–Schwarz in the Walsh basis gives $\sum_i y_i p(X_i) \le n\varepsilon\|p\|_2$, so $na < n\varepsilon\|p\|_2 + y^{\mathsf T} t_S$. If $2\varepsilon\|p\|_2 \le a$, then $y^{\mathsf T} t_S > na/2$. Otherwise $(y_i p(X_i))_- \le |t(X_i)|$ and coercivity give

$$\|t_S\|^2 \ge \frac{n}{4}\|p\|_2^2 > \frac{na^2}{16\varepsilon^2} \ge \frac{\delta n^2 a^2}{16 D_r} \ge s := \frac{\delta A_\delta \log(2D_r)}{16}\, na^2,$$

using $n \ge A_\delta D_r \log(2D_r)$. Thus $A_S \subseteq \{y^{\mathsf T} t_S > na/2\} \cup \{\|t_S\|^2 > s\}$. The first event has probability at most $\exp(-n^2a^2/(8y^{\mathsf T} Q_{SS} y)) \le \exp(-\delta n a^2/(16\tau_r))$. For the second, take $\vartheta = 1/(4\|Q_{SS}\|_{\mathrm{op}})$. Since $-\log(1-x) \le 2x\log 2$ for $0 \le x \le \frac12$, the Gaussian moment generating function satisfies $\log \mathbb{E} e^{\vartheta\|t_S\|^2} \le 2\vartheta \log 2 \operatorname{tr} Q_{SS} = 2\vartheta n\tau_r \log 2$. With Lemma 7.1,

$$\mathbb{P}\{\|t_S\|^2 > s\} \le \exp\Big(-\frac{s - 2n\tau_r \log 2}{4\|Q_{SS}\|_{\mathrm{op}}}\Big) \le \exp\Big(-\frac{s}{8 b_r \tau_r}\Big),$$

where the last step uses $s \ge 4n\tau_r \log 2$, valid for large d since $\tau_r \le A$ and $\log(2D_r) \ge \log(2d)$. Adding the two bounds proves (13). For $1 \le r \le d/64$ we have $\log(2D_r) \ge r\log(d/r) \ge 0.8\, rH_r$ and $rH_r \le 0.081 d$; hence $1 \le rH_r \le \sqrt{d\, rH_r}$ and $b_r \le 3C_\delta\sqrt{d\, rH_r}$, and the minimum in (13) is at least $\min\{\frac{1}{16}, \frac{1}{480 C_\delta}\}\min\{1, A_\delta\sqrt{rH_r/d}\}$. For large d the term $\log 2$ is at most half of the resulting main term, which accounts for the factor $\frac12$ in c_δ: indeed $na^2/\tau_r \ge na^2/A$, $\min\{1, A_\delta\sqrt{rH_r/d}\} \ge \min\{1, A_\delta\}/\sqrt{d}$ because $rH_r \ge 1$, and $n \ge \exp((\log(d+1))^3)$.

For (15), fix U_0 with $|U_0| = r+1$ and put $v = (\chi_{U_0}(X_i))_{i\le n}$. Every term of $Q = \sum_{|U|>r} \lambda_{|U|}\chi_U\chi_U^{\mathsf T}$ is positive semidefinite, so $Q_{SS} \succeq \lambda_{r+1} vv^{\mathsf T}$ and $\|Q_{SS}\|_{\mathrm{op}} \ge v^{\mathsf T} Q_{SS} v/\|v\|^2 \ge \lambda_{r+1}\|v\|^2 = n\lambda_{r+1}$. Also $\|Q_{SS}\|_{\mathrm{op}} \ge \operatorname{tr} Q_{SS}/n = \tau_r$. □

The proof of (15) uses only that $Q = \sum_{|U|>r} \lambda_{|U|}\chi_U\chi_U^{\mathsf T}$ with every $\lambda_q \ge 0$ and that Q has constant diagonal τ_r. So it holds for every covariance on $\mathcal X$ that depends only on Hamming distance, in particular for every kernel as in Section 3, and it uses neither the coercivity lemma nor the sample distribution.

Corollary 7.4 (The matching bound fails)**.** *In the setting of Proposition 7.3, for every $\Xi > 0$ there is $d_0(\Xi)$ such that, for $d \ge d_0(\Xi)$ and every n with $\exp((\log(d+1))^3) \le n \le e^{c_{\mathrm G} d}$, every sample in the event of Proposition 7.3 satisfies*

$$\frac{\mathcal D(S)}{n} \cdot \frac{\tau_r(b_r + rH_r)}{a^2} \;\ge\; \frac{\delta}{2}\min\Big\{\frac{C_\delta\sqrt{d\, rH_r}}{16}, \frac{A_\delta rH_r}{160}\Big\} \;>\; \Xi, \qquad r = r(n).$$

Hence the inequality

$$\frac{\mathcal{D}(S)}{n} \le \frac{\Xi\, a^2}{\tau_r\,(b_r + rH_r)} \tag{16}$$

holds with probability at most $4\delta < 1/2$ over the sample, and no sample event of probability greater than 4δ carries it.

Proof. Multiply (13) by $\tau_r(b_r+rH_r)/(na^2)$. Since $b_r+rH_r \ge b_r \ge C_\delta\sqrt{d\,rH_r}$ and $\log(2D_r) \ge 0.8\,rH_r$ (proof of Proposition 7.3), the main term is at least $\delta\min\{C_\delta\sqrt{d\,rH_r}/16,\ A_\delta rH_r/160\}$. It tends to infinity uniformly in n, because $\sqrt{d\,rH_r} \ge \sqrt{d}$ and $rH_r \ge r \ge (\log d)^2/4$ (Lemma 7.1). The term coming from $\log 2$ is $\log 2 \cdot \tau_r(b_r + rH_r)/(na^2) \le Cd/n$, because $\tau_r \le A$, $rH_r \le 0.081d$ and $b_r \le 3C_\delta\sqrt{d\,rH_r}$; it tends to zero. So for large d the left side is at least half the main term and exceeds Ξ. If (16) held on a sample event of probability greater than 4δ, that event would meet the event of Proposition 7.3, a contradiction. □

The constant Ξ may depend on δ, a, L and the gains, but not on d. Even a Ξ that grows with d must be at least $c\min\{\sqrt{d\,rH_r}, A_\delta rH_r\}$, which is at least of order $(\log d)^3$ at the bottom of the range and of order d at linear cutoffs.

Corollary 7.5 (The head-bound route at the margin scale)**.** *On the event of Proposition 7.3, let $\bar{\mathcal{D}} \ge \mathcal{D}(S)$ be any upper bound on the cost, let $\rho = r/d$ with $r = r(n)$, and let $u \ge r$. Then:*

(a) $16\|Q_{SS}\|_{\mathrm{op}}\bar{\mathcal{D}}/n \ge 16c_\delta a^2\,(n\lambda_{r+1}/\tau_r)\min\{1, A_\delta\sqrt{rH_r/d}\}$, *and $4\lambda_{r+1}\bar{\mathcal{D}}$ is at least a quarter of the right side;*

(b) $16b_r\tau_r\bar{\mathcal{D}}/n \ge 16c_\delta C_\delta a^2\min\{\sqrt{d\,rH_r}, A_\delta rH_r\} \ge 16c_\delta C_\delta a^2\rho\, dH_r$;

(c) for every $b > 0$,

$$\frac{u}{bd}\Big(8\tau_r + 16b_r\tau_r\frac{\bar{\mathcal{D}}}{n}\Big)^{1/2} \;\ge\; \frac{a}{b}\cdot 4\sqrt{c_\delta C_\delta}\,(\rho^3 dH_r)^{1/2}.$$

Proof. (a) follows from (15) and (14), since $\bar{\mathcal{D}} \ge \mathcal{D}(S)$. (b) follows from (14) and $b_r \ge C_\delta\sqrt{d\,rH_r}$, together with $\min\{\sqrt{d\,rH_r}, A_\delta rH_r\} \ge rH_r = \rho\, dH_r$, which holds because $rH_r \le d$ and $A_\delta \ge 1$. (c) follows from (b) and $u/d \ge \rho$. □

What this shows and does not show. Proposition 7.2 is about a proof technique, not about sample complexity, and it refutes the normalization [1, eq. (42)]. Proposition 7.3 is sharper. Its proof splits the statistic $\sum_i y_i G(X_i) > na$ into a low-degree part, which coercivity controls, and the high-degree part $y^{\mathsf{T}}t_S$. The prior variance $y^{\mathsf{T}}Q_{SS}y$ of the latter is of order $n\tau_r$ on typical samples, not $n\|Q_{SS}\|_{\mathrm{op}}$, because parity is orthogonal to every Walsh character of degree below d. The resulting lower bound exceeds that of Proposition 7.2 by a factor of order $\min\{\sqrt{d\,rH_r}, A_\delta rH_r\}$, which tends to infinity. The factor $\min\{1, A_\delta\sqrt{rH_r/d}\}$ in Proposition 7.3 is of order one when $\log n \asymp d$; at $n = \exp((\log(d+1))^3)$ the cutoff has $rH_r = (1+o(1))(\log d)^3$, and the factor is of order $A_\delta(\log d)^{3/2}/\sqrt{d}$. So Proposition 7.2 is not sharp, and no upper bound on the cost matches it: by Corollary 7.4, for every fixed Ξ, all large d and every n in its range, the inequality (16) fails on the event of Proposition 7.3, hence with probability at least $1 - 4\delta$.

Nor can any upper bound on the cost make the head-bound argument of [1], with Lemma 7.1, work at the margin scale a at any fixed $\rho > 0$. The clipped-parity lemma at scale a only needs the posterior's low-degree part to be $O(a)$ in L^2, not small, because it is multiplied by $u/d \approx \rho$ [1, eq. (24)]. The entropy bound at the training inputs and coercivity, [1, eqs. (41) and (43)], control that part by $\mathbb{E}^*\|G_{\le r}\|_2^2 \le 8\tau_r + 16\|Q_{SS}\|_{\mathrm{op}}\mathcal{D}/n$; this is the argument of [1, eq. (45)], which then

inserts that paper's Gram bound [1, eq. (44)]. By Corollary 7.5(a), whatever upper bound on $\mathcal{D}$ is inserted, the right side is at least $16c_\delta a^2(n\lambda_{r+1}/\tau_r)\min\{1, A_\delta\sqrt{rH_r/d}\}$, and the entropy estimate $2\tau_r + 4\lambda_{r+1}\mathcal{D}$ of the band $(r,u]$ is at least a quarter of that. With the bound $b_r\tau_r$ of Lemma 7.1 in place of $\|Q_{SS}\|_{\mathrm{op}}$, the head estimate is at least $16c_\delta C_\delta a^2\min\{\sqrt{d\,rH_r}, A_\delta rH_r\}$ (Corollary 7.5(b)). The resulting bound on $|\mathbb{E}^*\mathbb{E}_x f\,\mathrm{sign}\,G|$ is then at least $4(c_\delta C_\delta\rho^3 dH_r)^{1/2}$ (Corollary 7.5(c) with $b=a$), which exceeds one at every fixed $\rho>0$ once d is large. With the exact norm $\|Q_{SS}\|_{\mathrm{op}}$ the same happens if $n\lambda_{r+1}/\tau_r\to\infty$ at fixed ρ. That is a question about the spectrum alone: it holds if $\theta_r = \binom{D}{r+2}\lambda_{r+1}/\tau_r \ge c/\sqrt{r}$, which we observe numerically for the depth-2 tanh kernel at three gains and $d\le 3200$ (Section 8) but have not proved. If that ratio were only of order $1/r$, then $n\lambda_{r+1}/\tau_r$ would stay bounded at fixed ρ, and this argument would not exclude the exact-norm estimate. So an $\exp(\Omega(d))$ proof by the head-bound argument at the margin scale needs, with the bound of Lemma 7.1 (and with the exact norm too, if $\theta_r\ge c/\sqrt{r}$), a bound on the posterior's low-degree part that does not pass through $\|Q_{SS}\|_{\mathrm{op}}\mathcal{D}/n$, such as the route below.

These estimates concern the margin scale. If the clipped-parity lemma is applied at a scale $b>a$ instead, as [1] does at the amplitude scale of [1, Lemma 7.5], the lower bound above is multiplied by a/b (Corollary 7.5(c)). It still exceeds one if $b\le 4a(c_\delta C_\delta\rho^3 dH_r)^{1/2}$, so at fixed ρ for every $b=o(\sqrt{d})$; but for b of order $\sqrt{d}$ it tends to zero as $\rho\to 0$, so it is no obstruction there. With Lemma 7.1, such a version of the argument therefore needs an amplitude bound $\mathbb{E}^*\mathbb{P}_x\{|G(x)|\le b\} = o(1)$ at a scale b of order at least $\sqrt{d}$. [1, Lemma 7.5] does not give one at the standard scaling, and we neither prove nor exclude one. The only upper bound on the cost we have is the bump bound $e^{C\sqrt{d}}$.

A second sufficient route avoids the adaptive cutoff. It would suffice to show that, for a small fixed α,

$$\mathbb{E}^*\|G_{\le\alpha d}\|_2^2 \le C \quad\text{and}\quad \mathbb{E}^*\|G_{>\alpha d}\|_2^2 = o(1),$$

with C independent of α. By the clipped-parity lemma at scale a, $|\mathbb{E}_x f\,\mathrm{sign}\,G| \le \frac{\alpha}{a}\|G_{\le\alpha d}\|_2 + \|G_{>\alpha d}\|_2/a$, so it is enough that $\alpha\sqrt{C}/a \le 1/4 - o(1)$. By the Laplacian form [1, eq. (23)] of the clipped-parity lemma, the constant term may be excluded from the first condition. The second condition holds for $n\le e^{c_\alpha d}$ with $c_\alpha < \min\{\mathsf{h}(\alpha)/2, 0.09\}$, on the event of Lemma 5.1, which then has probability at least $1-e^{-c'd}$. Indeed Lemma 5.3 gives $\mathbb{E}^*\|G_{>s}\|_2^2 \le 2\tau_s + 4\lambda_{s+1}\mathcal{D}$, and Lemmas 2.6 and 5.2 make the right side $O(T(s)) + e^{C\sqrt{d}}n/\binom{D}{s+2}$ at odd $s\approx\alpha d$. The first condition is an upper bound on the posterior's low-degree amplitude, and we do not know how to prove it. The entropy bound gives only $\mathbb{E}^*\|G\|_2^2 \le 2A + 4\lambda_0\mathcal{D} = O(1+\mathcal{D}/d)$, which is $O(1)$ only if $\mathcal{D}=O(d)$; by Proposition 7.3, with probability at least $1-4\delta$, $\mathcal{D}/d \ge c\,n/d^{3/2}\to\infty$ uniformly for $\exp((\log(d+1))^3) \le n \le e^{c_{\mathrm{G}}d}$.

Heuristic picture (heuristic; not used above). For $\exp(d^\epsilon)\le n\le e^{c_{\mathrm{G}}d}$, Proposition 7.3 and the bump construction place the conditioning cost per sample between $c_\delta(a^2/\tau_r)\min\{1, A_\delta\sqrt{rH_r/d}\}$ and $e^{C\sqrt{d}}$; the lower end is superpolynomial in d. The mechanism behind the lower bound is coercivity together with the sign-aligned statistic. At the training inputs parity cannot be followed by a low-degree polynomial, so a low-degree part that carried the training margins would force a large high-degree part there; and a high-degree part that carries them by itself costs order na^2/τ_r, because parity is nearly orthogonal to the top eigenvectors of Q_{SS}. This holds whatever the posterior does: on the event of Proposition 7.3 the cost exceeds the bound of Proposition 7.2 by a factor tending to infinity. The bounds on the cost do not decide whether the posterior's low-degree part stays of order a, as the clipped-parity lemma at scale a needs, or whether the posterior uses large-amplitude low-degree thresholds that approximate parity. We have not ruled out the second behaviour, and the numerics below are too small to test it.

8 Numerical illustrations

All scripts and outputs are in `numerics/` next to this paper. They are illustrations, not evidence about asymptotic exponents.

Spectra (`spectrum.py`, `spectrum.txt`). We compute the composite coefficients c_m from Lemma 4.1 for the unit-gain network at $L = 2, 3, 4$ and for the specified-gain network at $L = 3$, for $m \le 600$. We then compute the cube eigenvalues through the walk representation $a_j = \sum_m c_m p_m(j)$, which involves no cancellation, at $d = 20, 40, 80, 160$. At $d = 10$ the eigenvalues agree with a direct Gauss–Hermite evaluation of the layer recursion followed by a Krawtchouk transform. The checks confirm positivity, monotonicity and pairing (Lemma 2.5), and the inequalities of Lemma 2.6(b),(c) and (5). Over the computed range, the level masses $\binom{d}{q}\lambda_q$ of the tanh kernels decay much faster than those of the arcsine kernel, whose level masses decay like $q^{-3/2}$ at $L = 2$ [1, Lemma 3.2]. Table 1 records representative values.

kernel	d	$q=1$	$q=3$	$q=9$	$q=17$	$q=33$	$q=65$	$v_{\min}$
tanh unit gain $L=2$	80	-0.04	-1.28	-3.26	-5.20	-8.86	-22.30	-15.3
tanh unit gain $L=2$	160	-0.03	-1.26	-3.19	-4.95	-7.77	-13.18	-23.3
tanh unit gain $L=3$	80	-0.05	-1.15	-3.06	-5.04	-8.78	-22.25	-15.3
tanh unit gain $L=3$	160	-0.05	-1.14	-2.98	-4.78	-7.67	-13.13	-23.3
tanh specified gain $L=3$	80	-0.07	-1.03	-2.67	-4.38	-7.82	-20.95	-13.9
tanh specified gain $L=3$	160	-0.06	-1.02	-2.60	-4.15	-6.76	-11.89	-21.3
arcsine $L=2$	80	-0.20	-0.98	-1.74	-2.17	–	–	-1.8
arcsine $L=2$	160	-0.20	-0.98	-1.73	-2.15	-2.60	–	-2.4

Table 1: $\log_{10}$ of the level mass $\binom{d}{q}\lambda_q/A$ of the normalized kernel $F((1 + x \cdot y)/(d + 1))$, and $\log_{10}$ of the smallest eigenvalue of the matrix $F((1 + x \cdot y)/(d + 1))_{x,y}$. Unit gain means $\sigma_\ell = 1$ for all ℓ; specified gain is the family of [2, Section 7] at $L = 3$. The computation truncates the slowly decaying arcsine coefficients at $m = 600$ (omitted mass 0.021), so arcsine entries are shown only where the omitted terms are negligible ($q \le 17$ at $d = 80$, $q \le 33$ at $d = 160$), and the arcsine $v_{\min}$ is only a lower bound. All checks listed in Section 8 passed for every row.

The share of the first level pair (`theta.py`, `theta.txt`). For large d the fast decay in Table 1 cannot continue at a fixed geometric rate up to linear degree: by Lemma 4.3(a) and Corollary 4.2, every level mass $\binom{d}{q}\lambda_q$ with $q \le 0.49d$ is at least $e^{-C\sqrt{d}}$, with C depending on the depth and gains, whereas a fixed factor below one per level would make it e^{-cd} near $q = 0.49d$. For odd F and odd r let

$$\theta_r = \frac{\binom{D}{r+2}\lambda_{r+1}}{\tau_r} = \frac{W_{r+2}}{\sum_{j \ge r+2,\ j \text{ odd}} W_j} \in (0, 1],$$

the share of the first level pair above r in the tail variance τ_r (see (12) and Lemma 2.6(b)). For the depth-2 tanh kernel we computed θ_r at $d = 160, 200, 400, 800, 1600, 3200$, with the coefficients of Lemma 4.1 and the exact walk recursion. Over all odd $r \le 0.065d$, $\theta_r\sqrt{r}$ lies in $[0.804, 3.06]$ for $\sigma_1 = 1$, in $[0.945, 5.43]$ for $\sigma_1 = 0.5$ and in $[0.610, 1.63]$ for $\sigma_1 = 2$. In each case the minimum is at $r = 1$, and the maximum is at the largest r and still increases with d: from 1.88 to 3.06 ($\sigma_1 = 1$), from 2.54 to 5.43 ($\sigma_1 = 0.5$) and from 1.20 to 1.63 ($\sigma_1 = 2$) between $d = 160$ and $d = 3200$. So in the computed range $\theta_r \ge c\, r^{-1/2}$, with $\theta_r\sqrt{r} \ge 0.61$ in all three cases, while the data do not settle

whether $\theta_r \le C\, r^{-1/2}$; for the arcsine kernel $\theta_r \asymp 1/r$ [1, Lemma 3.2]. Correspondingly, the ratio W_{j+2}/W_j of consecutive pair masses at $j \approx d/4$ is 0.48 at $d = 160$ and grows to 0.62, 0.78 and 0.84 at $d = 400$, 1600 and 3200 ($\sigma_1 = 1$), so in the computed range the decay per level seen at $d \le 160$ weakens as d grows. We prove neither bound on θ_r. The lower bound $\theta_r \ge c\, r^{-1/2}$ enters the exact-norm form of the obstruction in Section 7; the upper bound would be needed by the variant of the conditional route mentioned in Remark 6.18.

Sign-only posteriors (`signgp_parity.py`, `signgp_parity.txt`). We run exact Hamiltonian Monte Carlo for the orthant-truncated Gaussian, using the validated sampler of [3], for full parity at $d = 10, 12$. The kernels are tanh unit gain at $L = 2, 3$, tanh specified gain at $L = 3$, and the arcsine kernel at $L = 2$. Table 2 gives the confidence-failure fraction. In this range all four kernels fail on most unseen inputs (at least 71% of them in every cell). The whole-cube fraction falls mainly because more inputs are observed. The tanh kernels fail somewhat less often than the arcsine kernel, and the three tanh kernels are nearly indistinguishable. At these dimensions the numbers say nothing about the exponent.

$d = 10$, $n = 100, 200, 400, 800$				
tanh unit gain, $L = 2$	0.71 (0.78)	0.62 (0.75)	0.53 (0.79)	0.41 (0.90)
tanh unit gain, $L = 3$	0.71 (0.78)	0.62 (0.76)	0.54 (0.79)	0.41 (0.90)
tanh specified gain, $L = 3$	0.71 (0.79)	0.64 (0.78)	0.55 (0.81)	0.42 (0.92)
arcsine, $L = 2$	0.77 (0.85)	0.75 (0.91)	0.64 (0.94)	0.45 (0.98)
$d = 12$, $n = 200, 400, 800, 1600$				
tanh unit gain, $L = 2$	0.71 (0.74)	0.65 (0.71)	0.61 (0.74)	0.55 (0.82)
tanh unit gain, $L = 3$	0.71 (0.75)	0.65 (0.72)	0.62 (0.75)	0.55 (0.82)
tanh specified gain, $L = 3$	0.72 (0.76)	0.67 (0.74)	0.63 (0.77)	0.56 (0.83)
arcsine, $L = 2$	0.82 (0.87)	0.80 (0.88)	0.76 (0.92)	0.64 (0.95)

Table 2: Sign-only learner for full parity: confidence-failure fraction $F = \mathbb{P}_x\{q_S(x) \le 2/3\}$ over the whole cube (observed inputs count as successes), and in parentheses over unseen inputs only; mean over 3 seeds of iid uniform training inputs, columns in the order of the listed n. Two exact-HMC chains of 1500 draws after 300 burn-in; the largest difference in F between chains over all runs is 0.006.

9 Status summary

Statement	Status
Width limit to a nondegenerate GP, both learners	Proved (Prop. 2.4)
Sign-only learner, every training set, every activation/depth/gain: $\exp(\Omega(\sqrt{d}\log d))$ for full parity, d^k for fixed k-sparse parity	Proved (Thm. 3.2, Cor. 3.4)
Margin learner with readout exponent $b < 1$, tuple $(0, 1, \dots, 1, b)$: same bounds	Proved (Remark 3.5)
Margin learner, tanh, typical samples: $\exp(\frac{1}{20}\sqrt{d}\log d)$	Proved (Thm. 5.5)

Statement	Status
Same for odd activations whose first-layer coefficients are at least $e^{-o(\sqrt{m}\log m)}$, with $c_1(q_\ell) > 0$ at every later layer	Proved (Remark 5.6)
Restriction lemma: the agreement of a degree-u function with parity depends logarithmically on its amplitude	Proved (Lemma 6.1, Cor. 6.2)
Two-step regularity $c_{m+2} \geq \eta c_m$ of the tanh dual coefficients at every depth; $\tau_r \leq (1 + 4/\eta)\tau_{r+2}$	Proved (Lemmas 6.3–6.5)
Intrinsic-dimension bound $\mathbb{E}\|Q_{r,SS}\|_{\mathrm{op}} \leq (e-1)n\lambda_{r+1} + \tau_r \log(\tau_r/\lambda_{r+1})$	Proved, citing Lieb's concavity theorem (Prop. 6.6, Lemma 6.7)
Gram lower bound $\|Q_{SS}\|_{\mathrm{op}} \geq \max\{n\lambda_{r+1}, \tau_r\}$ on every sample	Proved (eq. (15))
Sign-only learner, tanh, typical samples: $n \leq \exp(10^{-3} d \log\log d / \log d)$ leaves failure fraction $\geq 3/8$ w.p. $\geq 3/4$, for $d \geq d_0$ with d_0 astronomically large; the same for the margin learner with readout exponent $b < 1$	Proved conditionally on [1, Lemma 4.1], which the subcritical theorem [1, Theorem 1.1] also uses; unconditional for $n \leq n_0(d)$ (Thm. 6.11, Remarks 3.5 and 6.15)
Same for odd activations with two-step regularity of the first-layer coefficients and $c_1(q_\ell) > 0$ at every layer, e.g. erf	Proved, with the same condition (Remark 6.17)
Per-sample conditioning cost superpolynomial for $e^{d^\epsilon} \leq n \leq e^{c_{\mathrm{G}} d}$ (w.p. $\geq 1 - 2\delta$); the subcritical normalization [1, eq. (42)] fails	Proved conditionally on [1, Lemma 4.1] (Prop. 7.2)
Lower bound $c_\delta(a^2/\tau_r)\min\{1, A_\delta\sqrt{rH_r/d}\}$ on the cost per sample (w.p. $\geq 1 - 4\delta$), so no upper bound matches Prop. 7.2; with Lemma 7.1, the head-bound argument of [1] at the margin scale a cannot give $\exp(\Omega(d))$; with the exact Gram norm the same holds if $\theta_r \geq c/\sqrt{r}$; clipping at a scale of order $\sqrt{d}$ is not excluded	Proved conditionally on [1, Lemma 4.1] (Prop. 7.3, Cors. 7.4 and 7.5); the exact-norm part also needs $\theta_r \geq c/\sqrt{r}$, which is only observed numerically
$\exp(\Omega(d))$ for either learner at fixed depth	Open
$\exp(\Omega(d))$ for the sign-only learner from log-free one-sided coercivity at linear degree and a log-free tail-Gram bound	Extension suggested by the proof, conditional on two open estimates (Remark 6.18)
For tanh, the level-mass decay seen at $d \leq 160$ cannot continue at a fixed rate up to degree $0.49d$	Proved (Lemma 4.3(a), Cor. 4.2)
$\theta_r \geq c\, r^{-1/2}$ for the depth-2 tanh kernel ($\theta_r\sqrt{r} \geq 0.61$ in all tested cases, $d \leq 3200$); whether $\theta_r \leq C\, r^{-1/2}$ is not settled	Numerical (Section 8)

Statement	Status
Which of the two posterior behaviours in Section 7 occurs	Open (heuristic discussion only)
Margin learner for erf, or for sparse parity with $k \ll d/\log^2 d$	Open

References

[1] D. Vaintrob, K. Hänni, and OpenAI Codex. Exponential parity sample complexity below the mean-field scaling. This collection, `main-results/subcritical-exponential`, 2026. Lemma and equation numbers refer to its current PDF.

[2] D. Vaintrob, K. Hänni, and ChatGPT. Small circuits under Gaussian-process classification: what sample complexity follows from the actual posterior? This collection, `side-results/nngp-sample-complexity`, 2026.

[3] Claude. Which targets are hard for the subcritical learners? This collection, `works-in-progress/beyond-parity`, 2026.

[4] J. Aspnes, R. Beigel, M. Furst, and S. Rudich. The expressive power of voting polynomials. *Combinatorica* 14(2):135–148, 1994.

[5] I. S. Gradshteyn and I. M. Ryzhik. *Table of Integrals, Series, and Products*, 7th ed. Academic Press, 2007.

[6] A. Daniely, R. Frostig, and Y. Singer. Toward deeper understanding of neural networks: the power of initialization and a dual view on expressivity. *NeurIPS*, 2016.

[7] R. M. Neal. *Bayesian Learning for Neural Networks*. Springer, 1996.

[8] J. Lee, Y. Bahri, R. Novak, S. S. Schoenholz, J. Pennington, and J. Sohl-Dickstein. Deep neural networks as Gaussian processes. *ICLR*, 2018.

[9] A. G. de G. Matthews, J. Hron, M. Rowland, R. E. Turner, and Z. Ghahramani. Gaussian process behaviour in wide deep neural networks. *ICLR*, 2018.

[10] J. R. Lee. Spectral hypergraph sparsification via chaining. arXiv:2209.04539. Section 2.4, Lemma 2.12 and equation (2.20).

[11] E. H. Lieb. Convex trace functions and the Wigner–Yanase–Dyson conjecture. *Advances in Mathematics* 11:267–288, 1973.

[12] J. A. Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics* 12:389–434, 2012.

[13] J. A. Tropp. An introduction to matrix concentration inequalities. *Foundations and Trends in Machine Learning* 8:1–230, 2015.

[14] A. Prékopa. On logarithmic concave measures and functions. *Acta Scientiarum Mathematicarum* 34:335–343, 1973.